View-Based Knowledge-Augmented Multimodal Semantic Understanding for Optical Remote Sensing Images

Lilu Zhu, Xiaolu Su, Jiaxuan Tang, Yanfeng Hu, and Yang Wang, *Member, IEEE*

***Abstract*— Optical remote sensing (RS) images serve as a pivotal source of geographic information. Due to the continuous development of deep learning technology, the evolving demands for multisource optical RS of the public shifted from recognition and acquisition of explicit features to comprehension and application of the fine-grained semantics and relationships implied in images. To address this challenge, we propose a semantic-augmented approach integrated multiview knowledge graph for a comprehensive understanding of optical RS images (RSMVKF). The RSMVKF delves into the structured representations of external knowledge from different human-like cognitive views and further explores the discovery ability of high-level features on the basis of multiple modalities and granularities. Specifically, the RSMVKF consists of two stages. First, we guide a large language model (LLM) to condense relevant knowledge from lengthy external knowledge passages and generate a view-level knowledge graph (RS-VKG). Then, an asymmetric multimodal contrastive network model (RS-M2CL) is designed to investigate efficient semantic augmentation. In this way, two types of contrastive loss functions, cross-modal and cross-granularity, are adopted to improve the understanding of implicit knowledge. The experimental results demonstrate that the RSMVKF greatly improves several perception tasks and reasoning tasks with rich features in optical RS imagery. In particular, in perception tasks such as fine-grained object detection and k-nearest neighbor (KNN) retrieval, the RSMVKF yields enhancements of 6.7% and 8.1%, respectively. In addition, in knowledge-driven reasoning tasks such as RS image captioning (RSCP), RS visual grounding (RSVG), and RS visual question answering (RSVQA), the RSMVKF demonstrates superior performance with margins of 8.9%, 5.3%, and 11.4%, respectively.**

***Index Terms*— Fine-grained semantics and relationships, multimodal contrastive network, multiview knowledge graph, optical remote sensing (RS) images, semantic-augmented approach.**

I. Introduction

With the rapid development of air–ground–space–sea integrated Earth observation technology and satellite–terrestrial integrated networks, an all-weather, all-round, and multiscale observation capability has been established, which has led to the continuous transmission of vast remote sensing (RS) images to the ground for analysis. RS technology leverages data fusion and mining methods to discern temporal, spatial, and spectral correlations between geographic entities and to investigate latent information in RS images. This provides humanity with the opportunity to perceive, understand, and reform the Earth in a novel manner. To date, optical RS, a principal branch of RS technology, has been used in a wide variety of applications, such as agriculture, land planning, smart cities, and disaster monitoring. Meanwhile, the evolving demand for multisource optical RS of the public has shifted from the recognition and acquisition of explicit features to the comprehension and application of the fine-grained semantics and relationships implied in images and has led to the emergence of numerous optical RS image analysis methods. Most existing methods follow a unified image learning paradigm that relies on substantial computational resources. In these methods, a deep neural network (DNN) comprising multiple layers and numerous parameters is constructed for feature extraction; this network is subsequently fine-tuned to project visual features from a common space into a semantic-specific space corresponding to various downstream tasks, such as scene classification [1], [2], [3], semantic segmentation [4], [5], change detection [6], [7], [8], and object detection [9], [10]. However, these methods remain at the recognition level, focusing on the analysis and modeling of shallow visual features to achieve the inductive representation of large-scale optical images. These models suffer from insufficient object-level abstraction, cannot generalize implicit semantic features and relations, and are unsuitable for the application of large-scale, high-resolution optical RS images [4], [6], [7], [11].

Recently, multimodal contrastive learning models (MCLMs) [12], [13], [14] and visual-language models (VLMs) [15], [16], [17], [18], [19], [20] have achieved outstanding performance in feature representation and semantic extraction for natural images, with applications in image–text retrieval [21], [22], visual question answering (VQA) [23], [24], and image captioning [25], [26]. The success of these methods in semantic extraction has garnered significant attention in the field of RS. Researchers attempt to apply these methods in RS visual-language tasks to enhance the semantic learning capabilities of models by extracting and integrating global visual features and text features [27], [28], [29], [30], such as RS visual question answering (RSVQA) [23], [24] and

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Lilu Zhu, Xiaolu Su, Jiaxuan Tang, and Yang Wang are with Suzhou Aerospace Information Research Institute, Suzhou 215123, China (e-mail: zhull@aircas.ac.cn; suxl300371@aircas.ac.cn; tangjx@aircas.ac.cn; wangyang003839@aircas.ac.cn).

Yanfeng Hu is with the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100094, China (e-mail: huyf@aircas.ac.cn).

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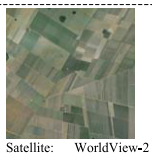
RS image captioning (RSCP) [25], [26]. Nevertheless, most VLMs trained on natural image–text always fail to account for the unique characteristics of RS images. In particular, these models tend to overlook the semantic complexity, confusability, and spatial relationships inherent in optical RS images. For example, in the optical RSCP task, although existing VLMs such as Blip-2 [15], InstructBLIP [16], MiniGPT-4 [17], LLaVA [18], Qwen-VL [19], and Gemini [20] can also provide general captions on the basis of the salient features of images, a significant gap remains between these outputs and human-like cognition. We leverage general multimodal VLMs (MVLMs) to generate image captions via zero-shot in-context learning to evaluate the reasoning ability of general MVLMs for optical RS images. Specifically, in Fig. 1, we present a comparative analysis of general state-of-the-art (SOTA) MVLMs for RSCP to evaluate their reasoning ability on the basis of multisource optical RS images. This analysis reveals the limitations of MVLMs in three main aspects. First, current MVLMs are deficient in their ability to discern complex semantics in optical RS images and thus fail to supply high-integrity and accurate RSCP results, as illustrated in Fig. 1(a). For example, in Fig. 1(a.1), the MVLMs provide a simplistic description of a lake's shape, neglecting the recognizable feature of the inverted C shape. Similarly, in Fig. 1(a.2), none of the MVLMs provide a comprehensive captioning result containing the arched steel girders. In Fig. 1(a.3), most MVLMs provide coarse-grained semantic features at a superficial visual level, lacking the precise information of the farmland category. Moreover, Fig. 1(a.4) illustrates that the MVLM does not indicate the location of the city as being at the confluence of two rivers. Second, existing MVLMs exhibit a limited ability to distinguish between RS images with similar visual characteristics. For example, as shown in Fig. 1(b), there are certain visual similarities between the RS images of the desert category and bare land in terms of color and texture, raising the requirements for the model's ability to understand the RS image. Consequently, some MVLMs misjudge when discriminating between optical RS images of these two categories, as shown in Fig. 1(b.2) and (b.3). Third, current MVLMs often fail to accurately describe the spatial relationships and semantic correlations of features within optical RS images, as depicted in Fig. 1(c). For example, Fig. 1(c.1) presents inadequate descriptions of the spatial relations between aircraft and buildings; Fig. 1(c.2) demonstrates a lack of discrimination regarding spatial relations among circular buildings, green trees, and rivers; and Fig. 1(c.3) offers an unsatisfactory description of the spatial distributions of circular commercial buildings, skyscrapers, and roads. Therefore, it is challenging for existing MVLMs to supply comprehensive descriptions and high-level semantic representations for optical RS images. Two fundamental constraints warrant consideration. On the one hand, the inherent characteristics of multisource optical RS images present significant challenges. Unlike natural images, optical RS images have multitemporal and multiscale characteristics, where the same geographic entities always show noticeable differences in visual features. Additionally, there are enormous geographic entities with significant size variations in complex optical RS images, leading to unignorable intraclass heterogeneity and interclass homogeneity. Finally, complex relationships exist among entities in images in terms of semantics, time, and space, which are defined differently in diverse specific contexts of optical RS images. On the other hand, current MVLMs lack a deep understanding and analysis of RS images, which impairs their ability to uncover the fine-grained semantics and complex patterns of relations among objects embedded within optical RS images. Overall, in the domain of multisource optical RS images, enhancing model semantic understanding by solely optimizing model architecture with a focus on visual features hits a bottleneck that is challenging to overcome.

Researchers observe that in the process of object recognition and understanding of RS images, individuals do not rely solely on the visual information provided by the images; instead, they often draw upon their existing knowledge and experience [27], [31], [32], [33], [34], [35], [36]. This insight offers a new approach to refined semantic analysis and relation extraction in RS images, suggesting that the desired results can be achieved by incorporating external knowledge into the model training process. In recent years, many RS multimodal learning methods with external knowledge have emerged and demonstrated noteworthy advancements. Nevertheless, these methods also present certain limitations. For example, Ji et al. [27] constructed a knowledge system based on valuable concepts but focused on a single view, neglecting the interactions between knowledge from multiple views, such as the spatial relation view (SRV). Liu et al. [31] incorporated textual information as a knowledge source to guide visual feature learning. Mi et al. [32] utilized external knowledge sources to construct an RS knowledge graph (RSKG). However, their analysis is limited to the literature [34] and the knowledge covered in ConceptNet. Furthermore, the knowledge extracted in [34], [35], and [36] is not generalized, making it applicable only to specific domains. While fine-grained semantic understanding of RS images typically requires specialized domain-specific knowledge, most of the knowledge introduced by existing methods is general conceptual knowledge, such as Wikipedia[1] and ConceptNet.[2] These textual categories of knowledge are extensive in scope but often exhibit a paucity of depth. Lengthy external RS knowledge passages are often derived from a variety of sources, including common-sense knowledge bases (e.g., Wikipedia[1] and ConceptNet[2]), basic geographic information databases (e.g., OpenStreetMap[3] and Geonames[4]), and optical RS metadata files[5,6] (e.g., Landsat-8, GF-2, MODIS, and Sentinel-2). Therefore, difficulties are encountered in pursuing the consistency of structure and content. Inappropriate integration would reduce the accuracy of the analysis results or even lead to erroneous conclusions. The urgent challenges currently faced are how to condense relevant knowledge from lengthy external RS knowledge passages and

[1]Wikipedia: https://www.wikipedia.org/
[2]ConceptNet: https://conceptnet.io/
[3]OpenStreetMap: https://www.openstreetmap.org/
[4]Geonames: https://www.geonames.org/
[5]https://www.gscloud.cn/
[6]https://earthexplorer.usgs.gov/

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(a) Complexity

	Optical RS Data	Model Name	Model Response
(1)	 Satellite: WorldView-3 Time: 2020-06-04 Resolution: 1.2m	Blip-2	The image is **a lake** in the suburbs of a city.
		MiniGPT-4	This is a satellite image of **a lake** in a residential area. The lake is surrounded by houses and trees. The water is clear and there are no boats or other watercraft in the lake.
		Qwen-VL-Plus	The image presents an aerial view of a residential area nestled around **two large lakes**, with houses and buildings scattered throughout.
		LLaVa-v1.6	The image presents an aerial view of a residential area with a large, curved lake at its center. **The lake's shape is irregular**, with multiple bays and inlets, and it appears to be a significant feature of the landscape. Surrounding the lake are numerous houses, which are densely packed and vary in size and design.
		Gemini-1.0-pro-vision	The image you sent me is an aerial view of a freshwater lake. The lake is a light blue color and has **a slightly irregular, winding shoreline**. There are trees ringing the lake, and houses with dark roofs can be seen nestled among them. The properties lining the lake appear to be quite spacious.
(2)	 Satellite: World View-2 Time: 2020-03-10 Resolution: 0.31m	Blip-2	The image shows **a bridge** over a canal.
		MiniGPT-4	The image shows **a bridge** over a river in a city. The bridge has several cars driving over it, and there are buildings on either side of the river. The river is surrounded by trees and other vegetation.
		Qwen-VL-Plus	The image captures an aerial view of a bridge, its concrete structure spanning across two bodies of water. **The bridge is adorned with three arches**, each varying in size and positioned at different points along its length.
		LLaVa-v1.6	The image shows an aerial view of **a bridge** crossing a body of water, with a road leading up to it and a car on the bridge. The water appears calm and the surrounding area is landscaped with greenery and concrete structures.
		Gemini-1.0-pro-vision	The image shows an aerial view of a long, multi-lane bridge spanning a winding river. **The bridge has four towers supporting its cables**, and there is traffic flowing steadily across it in both directions.
(3)	 Satellite: WorldView-2 Time: 2017-02-28 Resolution: 0.5m	Blip-2	The image is a view of **a field of green and brown fields**.
		MiniGPT-4	The image shows **a large, green field** with a few trees scattered around it. The field is divided into several sections, each with a different crop. Some of the crops are tall and green, while others are shorter and yellow. There are also some small patches of brown and yellow in the field.
		Qwen-VL-Plus	A satellite view of an area with **many fields and buildings**.
		LLaVa-v1.6	The image shows a satellite view of a field with a pattern of green and brown squares, indicating **a crop rotation system**.
		Gemini-1.0-pro-vision	The image shows an aerial view of a patchwork of **green and brown farm fields**.
(4)	 Satellite: WorldView-3 Time: 2018-10-28 Resolution: 0.31m	Blip-2	The image shows **the city** of shanghai, china.
		MiniGPT-4	The image shows a satellite view of **a city with a river running through it**. The river is surrounded by buildings and trees on both sides. The city has a mix of modern and traditional architecture, with some high rise buildings and some low rise buildings. The city appears to be large and has a lot of traffic on the roads.
		Qwen-VL-Plus	The image is an aerial view of a cityscape. The city appears to be densely populated with buildings scattered throughout its area. There's also evidence of **water bodies within the city limits, possibly rivers or lakes**.
		LLaVa-v1.6	The image is an aerial or satellite view of a densely populated urban area. The cityscape is characterized by a network of roads and buildings, with **a river flowing through the center, dividing the city into two distinct halves**. The river is flanked by bridges, indicating a well-developed infrastructure for transportation.
		Gemini-1.0-pro-vision	The image you sent me is an aerial view of a city. **The city is divided by a winding river**, with bridges crossing at several points. The buildings in the city are mostly rectangular and brown, with some green spaces scattered throughout.

(b) Confusability

	Optical RS Data of Desert Category	Model Name	Model Response
(1)	 Satellite: WorldView-3 Time: 2018-10-28 Resolution: 0.31m	Blip-2	The image is of a sand dune. √
		MiniGPT-4	The image shows a sandy beach with waves crashing against the shore. The sand is smooth and shiny, with small ripples and waves visible in the water. The beach is long and stretched out. √
		Qwen-VL-Plus	The entire image is filled with sand, creating an impression of being on a beach or desert landscape where water has recently receded to reveal these patterns and ripples. The light beige color suggests it might be early morning or late afternoon when sunlight casts soft shadows across the surface. X
		LLaVa-v1.6	The image shows a close-up view of a textured surface that appears to be a type of fabric or material with a wavy, undulating pattern. The color is a uniform light beige or cream, and the texture is smooth with subtle variations in height, creating a sense of depth and movement. X
		Gemini-1.0-pro-vision	The image you sent is a close-up of sand. The sand is light brown and very fine-grained. There are ripples across the surface of the sand, likely caused by wind. √
(2)	 Satellite: WorldView-2 Time: 2019-12-12 Resolution: 0.5m	Blip-2	The image shows a sand dune with a lot of sand. √
		MiniGPT-4	The image shows a sandy terrain with some small dunes and a few rocks in the foreground. The colors are mostly brown and sandy, with some rocks and plants in the foreground. The image is quite detailed and shows a lot of texture and variation in the terrain. √
		Qwen-VL-Plus	This is a satellite view of a dirt road that cuts through some hills. There's also a small house visible near the bottom left corner of this picture. X
		LLaVa-v1.6	The image shows a close-up view of a textured surface that appears to be a natural formation, possibly a rock or a section of a cliff. The surface is characterized by a series of parallel ridges and valleys, creating a wavy pattern that resembles a cross-section of a riverbed or a desert landscape. X
		Gemini-1.0-pro-vision	The image you sent is an aerial view of a desert landscape. The ground is covered in brown sand with some windblown ridges. There is no vegetation visible in the image. √

	Optical RS Data of Bareland Category	Model Name	Model Response
(3)	 Satellite: WorldView-3 Time: 2020-10-03 Resolution: 0.5m	Blip-2	The image shows a large construction site. √
		MiniGPT-4	The image shows a satellite view of a large construction site in the middle of a desert. The site is surrounded by a large dirt road and several large trucks are parked on the side of the road. X
		Qwen-VL-Plus	This is a satellite view of a dirt road that cuts through some hills. There's also a small house visible near the bottom left corner of this picture. X
		LLaVa-v1.6	The image appears to be a satellite or aerial view of a landscape that has been significantly altered by human activity. The terrain is dominated by large, irregularly shaped excavations and earthworks, suggesting mining or construction activities. The ground is mostly bare, with patches of vegetation scattered across the area. √
		Gemini-1.0-pro-vision	The image you sent me is an aerial view of a dirt road winding through a desert landscape. The road is a reddish brown color and appears to be well-traveled. The surrounding landscape is arid and rocky, with sparse vegetation. X
(4)	 Satellite: WorldView-2 Time: 2020-02-20 Resolution: 1.2m	Blip-2	The image is of a construction site. √
		MiniGPT-4	The image shows a large construction site with a large pile of dirt and debris in the center. There are several vehicles parked on the street nearby, and a few people are walking around the site. The overall scene is one of construction and activity. √
		Qwen-VL-Plus	The image captures an aerial view of a construction site, where earth has been excavated to reveal layers beneath it. The ground is scattered with rocks and debris from the excavation process. A few vehicles can be seen parked on the right side of the photo, possibly belonging to workers or equipment operators. √
		LLaVa-v1.6	The image presents an aerial view of a construction site. The site is characterized by a large, open area with various excavations and piles of materials. There are several large mounds of earth, some of which appear to be freshly excavated, and others that seem to be in the process of being moved or sorted. The ground is mostly bare, with patches of grass and small trees scattered around the perimeter. √
		Gemini-1.0-pro-vision	The image you sent is an aerial view of a large construction site. The site is cluttered with various construction vehicles and materials. In the center, a yellow bulldozer is pushing a pile of dirt. There are also dump trucks, excavators, and stacks of pipes and lumber scattered around the site. A dusty road winds through the construction zone. √

Fig. 1. Optical RSCP results by various MVLMs with the instruction "Present a description of this image." (a) Semantic complexity of optical RS images. The red text represents the coarse-grained semantic features extracted by the model, while the green text indicates the fine-grained semantic features extracted by the model. (b) Confusability of optical RS images. (c) Spatial relations between objects in optical RS images. These optical RS images are obtained from the Esri World Imagery platform. The figure displays the satellite, acquisition time, spatial resolution, and location corresponding to each optical RS image.

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(c) Spatial relation

	Optical RS Data	Model Name	Model Response
(1)	 Satellite: WorldView-2 Time: 2021-11-25 Resolution: 0.31m	Blip-2	A large airport with many planes parked on the tarmac.
		InstructBlip	A group of planes parked at an airport.
		MiniGPT-4	The image shows an aerial view of an airport with several rows of parked airplanes on the tarmac. The airport appears to be quite large, with multiple terminals and runways. There are also several buildings and other structures visible in the background, including a control tower and a maintenance hangar.
		LLaVa-v1.6	The image is an aerial or satellite view of an airport apron area. There are several airplanes parked at gates or on the tarmac, with their wings and tails visible. The aircraft vary in size, suggesting a mix of commercial and possibly private or service planes. The tarmac is marked with lines and there are buildings and structures that are part of the airport infrastructure. The overall scene is orderly, with no visible activity, indicating a moment of calm or a period of low traffic at the airport.
		Gemini-1.0-pro-vision	The image you sent is an aerial view of a busy airport with many airplanes parked on the tarmac. There are long, parallel runways stretching out from the terminal buildings. Some airplanes are taxiing towards the runways, while others are lined up waiting to take off. The terminal buildings themselves are long and rectangular with jetways connected to some of the airplanes. There are also various buildings and structures scattered around the airfield, likely for maintenance and storage.
		[Reference]	**Eight white planes** are parked **near** **a large building** in an airport.
(2)	 Satellite: WorldView-3 Time: 2018-03-03 Resolution: 0.31m	Blip-2	The image is of the chicago river.
		InstructBlip	A large building with a large circular roof.
		MiniGPT-4	The image shows a large building with a white exterior and a large glass dome on top. The building is surrounded by a large body of water, which appears to be a lake or a river. There are several trees and other vegetation around the building, and a few boats can be seen in the water. The building appears to be a large conference center or a museum, with several entrances and windows on the ground floor. The glass dome on top of the building is large and circular, with a series of smaller windows around the edge. The building appears to be made of concrete and steel, with a modern, sleek design.
		LLaVa-v1.6	The image presents an aerial view of a large, modern building complex situated on a peninsula surrounded by water. The architecture is contemporary, with a mix of flat and curved surfaces, and the building features a prominent circular structure at its center. The complex is landscaped with greenery and pathways, and there are several smaller structures and parking areas surrounding the main building. The water around the peninsula is calm, and the overall setting suggests a well-planned and possibly educational or research facility, given the layout and the presence of what appears to be a solar panel array on the roof.
		Gemini-1.0-pro-vision	The image you sent is an aerial view of the Shedd Aquarium in Chicago. It is a large, rectangular building with a curved roof. The roof is covered in solar panels. The aquarium is surrounded by water on three sides, with a narrow strip of land connecting it to the mainland. There are also several smaller buildings next to the aquarium, which may be part of the Shedd complex. The water is a blue-green color, and there are a few small boats visible in the distance.
		[Reference]	**A building** **surrounded** by some **green plants** are **near** a **river** .
(3)	 Satellite: GeoEye01 Time: 2021-12-10 Resolution: 0.31m	Blip-2	The image shows a city in china.
		InstructBlip	An aerial view of a city.
		MiniGPT-4	The image shows a bird’s eye view of a city with tall buildings and a busy street. The buildings are made of concrete and steel, with windows and balconies on the upper floors. The street is lined with cars and pedestrians, and there are trees and parks in the distance.
		LLaVa-v1.6	The image presents an aerial view of a densely populated urban area. The cityscape is characterized by numerous high-rise buildings, some of which are clustered together, while others are more spread out. The architecture varies, with some buildings having a more modern design, while others appear to be older or have a more traditional style. The roads and infrastructure weave through the city, connecting the buildings and creating a network of pathways. There are also areas of greenery, likely parks or gardens, interspersed among the buildings, providing a contrast to the urban environment.
		Gemini-1.0-pro-vision	The image you sent is indeed an aerial view of a city with tall buildings. The buildings vary in height and architectural style, but most are made of glass and metal with flat roofs. There is a cluster of skyscrapers in the center of the image, but there are also many mid-rise buildings. Reflecting the sunlight, the windows of the buildings create a mosaic of bright and dark squares. Interspersed among the buildings are green spaces, some with trees and some with landscaped lawns.
		[Reference]	**The circle commercial** is **opposite** to some **high rise buildings** on **either side** of **the road**.

Fig. 1. *(Continued.)* Optical RSCP results by various MVLMs with the instruction “Present a description of this image.” (a) Semantic complexity of optical RS images. The red text represents the coarse-grained semantic features extracted by the model, while the green text indicates the fine-grained semantic features extracted by the model. (b) Confusability of optical RS images. (c) Spatial relations between objects in optical RS images. These optical RS images are obtained from the Esri World Imagery platform. The figure displays the satellite, acquisition time, spatial resolution, and location corresponding to each optical RS image.

how to improve the understanding capability of RS multimodal learning models by integrating this relevant knowledge.

To overcome the aforementioned challenges, we focus on the structured representation of RS knowledge and leverage these representations to explore the discovery ability of high-level features integrated with current mainstream learning paradigms, such as DNNs and multimodal self-supervised learning, as illustrated in Fig. 2. We propose a semantic-augmented approach integrated multiview knowledge graph for a comprehensive understanding of optical RS images (RSMVKF). The RSMVKF delves into the structured representations of external relevant knowledge from different perspectives of human-like cognition and further explores the discovery ability of high-level features on the basis of multiple modalities and granularities. This approach consists of two stages. First, we guide the large language model (LLM) following the principles of human-like cognition to condense relevant knowledge from lengthy external knowledge passages and generate the view-level knowledge graph (RS-VKG). Second, we introduce an asymmetric multimodal contrastive network model (RS-M2CL), which includes an image encoder, a text encoder, a knowledge encoder, and a multimodal encoder, to investigate efficient semantic augmentation. In this stage, cross-modal and cross-granularity contrastive loss functions are adopted to improve the understanding capability of implicit knowledge. Our approach compensates for the shortcomings of conventional RS analysis methods in learning abstract semantics, relational semantics, and sparse semantics, and optimizes model performance in downstream perception and reasoning tasks. The contributions of this article are summarized as follows.

1) We propose a semantic-augmented approach to improve the understanding of implicit content in multisource optical RS images, which adopts a two-stage learning schema. In the first stage, we leverage the LLM to condense relevant knowledge from lengthy external RS knowledge passages as a view-level knowledge graph. In the second stage, an asymmetric contrastive network model is devised that integrates text, vision, and structured graphs, enabling a comprehensive semantic understanding.
2) To better align with human cognition for general knowledge and reduce reasoning hallucinations in downstream tasks, we first attempt to compress external RS knowledge passages into a distilled knowledge graph using the ability of LLMs to comprehend text. Notably, we further divide and condense external RS knowledge from four perspectives of human-like cognition, which lowers the entry bar of model execution.
3) To validate the effectiveness of the RSMVKF, we present an image–text paired dataset with lengthy external knowledge passages in the field of optical

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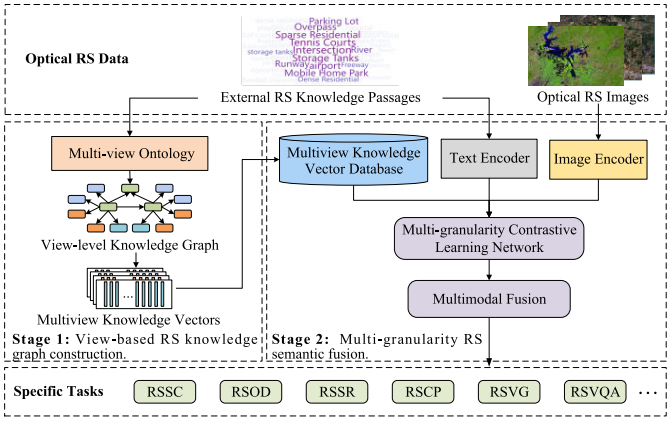


Fig. 2. Illustration of our proposed RSMVKF.

RS, RSVKG100H, which is obtained from filtered public RS datasets and generated by the pretrained world model. The experimental results demonstrate that the RSMVKF yields enhancements of 6.7% and 8.1% in fine-grained RS object detection (RSOD) and k-nearest neighbor (KNN) RS retrieval, respectively. In addition, in knowledge-driven reasoning tasks such as RSCP, RS visual grounding (RSVG), and RSVQA, the RSMVKF achieves superior performance with margins of 8.9%, 5.3%, and 11.4%, respectively.

II. Related Works

In this section, we review current studies of RS knowledge representation, RS multimodal learning, and RS multimodal learning with external knowledge.

A. RS Knowledge Representation

RS knowledge representation refers to the structured representation of RS-relevant knowledge condensed from lengthy knowledge passages. In recent years, significant progress has been made in knowledge-driven RS research [36], [37], [38], [39]. The primary methods of RS knowledge representation include knowledge graph-based representations [32], [40] and DNN-based representations [8], [57], [58], [59]. A knowledge graph, which encapsulates rich relational information, has the potential to represent complex semantic knowledge in the RS field. It encompasses both the attributes of entities and the relationships between them [32], [40]. The knowledge graph first proposed by Google in 2012 represents a network of knowledge comprising entities, attributes, and relations. Commonly used knowledge graphs include ConceptNet [41], WikiData [42], DBpedia [43], and Yago [44]. Recent studies have shown that models incorporating knowledge graphs demonstrate superior performance in a variety of complex visual understanding tasks [45], [46], [47]. Over the past few years, knowledge graphs have gradually played a crucial role in the intelligent interpretation of RS images [48]. Knowledge graphs covering RS prior knowledge have also been created [33], [37], [48], [49], [50], [51], [52], [53], [54], [55], [56]. For example, Li et al. [33] developed an RSKG that enhances the semantic representation of an RS scene, thereby enabling the reasoning and recognition of previously unobserved RS scenes. Fang and Yan [50] devised a knowledge graph based on the spatial relationships between objects of different land-cover scene categories, with the intention of assisting in multimodal RS scene classification (RSSC). Liu et al. [52] created a multisource knowledge graph for marine oil spill detection on the basis of organized information from various sources, including RS images, lengthy passages, and atmosphere–ocean models, to identify oil spills in RS images. Chen et al. [53] developed a multisource data-driven landslide prediction method containing a spatiotemporal knowledge graph integrated model that is capable of forecasting landslides in areas with limited samples. Hao et al. [37] constructed a knowledge graph integrating large-scale multisource heterogeneous data to measure the correlation between the surveying and mapping field and the RS field. Additionally, some studies have proposed methods to integrate knowledge graphs to improve the accuracy of RS image recommendations [56]. Adequate domain knowledge serves as the foundation for RSKG construction. However, the acquisition of such knowledge is frequently influenced by manual involvement from various experts, which requires a considerable investment of time and effort. Furthermore, the generalization and extension of RSKGs to accommodate the requirements of diverse downstream tasks in the field of RS present additional challenges.

B. RS Multimodal Learning

Multimodal learning establishes connections between visual and text modalities through joint learning on image–text sample pairs. It is typically pretrained via self-supervised approaches, as outlined in [60] and enables models to obtain common representations of images and texts. Contrastive learning [61], a principal method in self-supervised learning, learns effective representations by minimizing the distance between similar samples and maximizing the distance between disparate samples. Due to its exceptional performance in cross-modal alignment and zero-shot learning, contrastive learning has been widely adopted in multimodal learning, resulting in the development of foundational models such as SimCLR [62], CLIP [12], MoCo [63], and MoCo-v2 [64]. In addition, in the field of RS, numerous models based on contrastive learning have emerged [65], [66], [67], [68], [69], [70], [71], [72], [73], [74], [75]. GASSL [65] uses spatially aligned RS images to construct spatiotemporal samples utilized in MoCo-v2, incorporating an additional geographic location prediction pretraining task. RS-CLIP [67] acquires semantically aware visual representations on the basis of CLIP and incorporates pseudo-labeling techniques to address the challenges of zero-shot RSSC. DINO-MC [69] applies self-supervised learning with multisized views on the basis of the DINO [70] framework. TSC [66] uses self-attention and cross-attention modules to perform contrastive learning on the feature correlations of RS images from different views. GLCNet [68] integrates global style and local matching contrast learning modules and is applied to the semantic segmentation of RS images. SeCo [71] and CACo [72]

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employ contrastive learning to detect differences present in RS images captured at disparate timestamps. DCN [73] performs context-guided and detail-guided contrastive learning simultaneously to discover the interclass and intraclass features of RS images. MCRN [74] introduces a shared pattern transfer module based on pattern memory and combines domain alignment with contrastive learning for self-supervised pretraining, facilitating cross-modal semantic alignment in the RS. DUCH [75] employs contrastive learning that integrates a feature extraction module and a hashing module to increase the efficiency of cross-modal image–text retrieval. RSICCformer [76], a Transformer-based RS image change captioning model, contains a dual-branch transformer (DTE) consisting of a hierarchy of processing stages to capture and recognize multiple changes of interest. Prompt-CC [77] introduces prompt learning and the LLM into the RS image change captioning task and proposes a multiprompt learning strategy to effectively exploit the potential of the LLM for captioning.

C. RS Multimodal Learning With External Knowledge

In recent years, researchers have devoted increasing attention to integrating external expert or domain knowledge from various sources with RS models to increase their overall performance [27], [32], [33], [35], [37], [78], [79], [80], [81], [82]. For RS image learning, Li et al. [36] proposed a domain knowledge-guided deep collaborative fusion network that leverages multimodal RS indices as domain knowledge for supervision. This network implements an end-to-end domain knowledge reconstruction process to improve model performance in land-cover scene classification. Similarly, Hao et al. [37] introduced geographical locations associated with RS images as geographical knowledge to improve network performance in representation learning. Yan et al. [78] proposed a domain knowledge-guided self-supervised learning approach to improve model performance in RS change detection, incorporating RS indices throughout the training and inference processes.

Recent studies have also demonstrated significant progress in RS multimodal learning by incorporating external knowledge on various downstream tasks, such as RS image retrieval [27], [32], [79], [81], RSVQA [84], [85], and RS knowledge recommendation [57]. For example, Ji et al. [27] developed a knowledge-aided momentum contrastive learning framework for RS image–text retrieval, which introduces valuable concepts and learns discriminative representations to capture multilevel information from RS images. Mi et al. [32] proposed a text-guided knowledge transfer method that utilizes textual information with minimal semantic differences to narrow the semantic gap between image modality and text modality. Additionally, Mi et al. [79] devised a knowledge-aware cross-modal RS image–text retrieval method, which enriches text content by mining relevant information from an external knowledge graph to alleviate the information gap between texts and images for semantic alignment. Pan et al. [81] used RS knowledge from the literature [83] for RS scene recognition to guide the adaptive learning of image and text representations to enhance vision-language comprehension.

III. Methods

A. View-Based RSKG

The external RS knowledge required in a view-based RSKG encompasses the foundational knowledge used to support RS image upstanding in domain-specific applications, which consists of concepts, experiences, and principles related to real-world objects derived through abstraction and induction. In this section, we provide an efficient structured representation of external knowledge by constructing a view-level RSKG. This knowledge graph delineates the concepts, attributes, and their interrelations within the RS domain from four perspectives of human-like cognition. Typically, the ontology establishment should precede the knowledge graph construction. Here, we first define the view ontology on the basis of the RS ontology, which serves as the semantic conceptual model for view-level knowledge graphs. In this work, the view ontology includes classes, attributes, and relations among them. Classes within the RS domain can be subdivided and extended on the basis of physical entities, such as transportation, man-made structures, water bodies, vegetation, and ground cover. Each class encompasses various subclasses corresponding to specific objects. For example, transportation includes airplanes, vehicles, and boats, whereas man-made structures include buildings, bridges, and roads. These objects possess various attributes, such as name, shape, color, geographic location, and area. The RSKG typically contains rich relational information. Following the general principles of human-like cognition, the view ontology comprises four view types, including the object view (OV), the attribute view (AV), the spatial position view (SPV), and the SRV. As illustrated in Fig. 3, we condense and divide external RS knowledge from the perspective of human-like cognition and construct a multiview knowledge graph in terms of four views. The OV focuses primarily on the basic information of a specific object, including its name, type, and other identifiers. The AV emphasizes information that reveals the essential characteristics of an object, such as shape, color, area, function, width, and height. The SPV is concerned with the location and distribution characteristics of an object within a spatial context, including its geographic location and spatial distribution. Finally, the SRV examines the spatial relationships between a particular object and other objects, which can be further divided into spatial orientation relationships, such as Northwest, North, Northeast, East, and Southeast, and spatial topological relationships, including surrounding, intersecting, adjacent, and nearby.

First, we condense multiview knowledge containing entities and the relationships between them from external knowledge passages. We express the obtained entities and relationships as nodes and edges, respectively, in view-level knowledge graphs. SOTA approaches often apply LLMs to transform lengthy external knowledge passages into knowledge graphs, leveraging their contextual understanding and linguistic capabilities for entity and relationship discovery [86], [87], [88], [89], [90]. The earliest application of LLM for knowledge graph construction is commonsense transformer (COMET), introduced in 2019. COMET constructs a knowledge graph

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Relation types	① Object view	② Attribute view	③ Spatial position view	④ Spatial relation view	
				topological	directional
	name is	shape is	located in	surrounded by	north
	is component of	color is	on	intersect at	west
	is part of	height is	is distributed	next to	northeast
	belong to	area is	stop at	near	southwest
	has	is used to	over	around	…

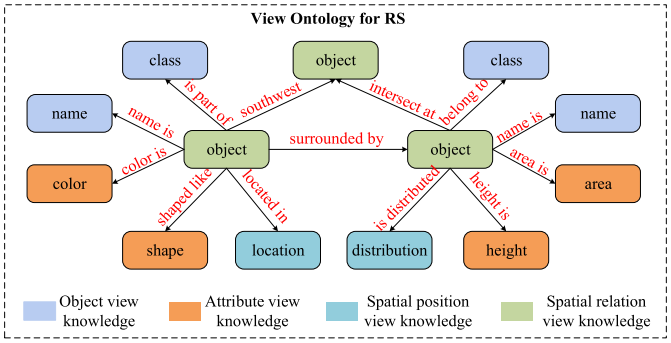


Fig. 3. View ontology for RS.

by generating tail entities via fine-tuned generative LLMs [87]. In recent years, researchers have explored the use of prompt engineering to fine-tune LLMs. By designing and optimizing input prompts and keeping pretrained parameters fixed, they can guide the model to generate information for knowledge graphs [88], [89], [90]. More recently, instruction fine-tuning has shown superior generative capabilities of LLMs with the provided instruction-following dataset, which enables the models to better understand and follow the task instructions [91], [92], [93]. Drawing on these insights, we employ a specified prompt template and instruction list for triplet extraction from four views to guide the LLM in generating knowledge triplets used to construct the view-level knowledge graph.

As illustrated in Fig. 4, we propose a view-based knowledge graph construction method, which guides the LLM to condense RS knowledge from four perspectives of human-like cognition from lengthy external knowledge passages and generates the view-level knowledge graph. First, for given RS images, we employ a generic VLM to generate an initial image caption. Then, we use an automated text processing toolkit to obtain relevant external knowledge in the field of optical RS from different sources, including common-sense knowledge databases, basic geographic information databases, and optical RS metadata files. For example, the classes and attributes of geographic entities can be obtained from open access platforms such as OpenStreetMap, and their name and location information can be obtained from GeoNames; the spatial resolution can be obtained from the elements in optical RS metadata files such as "GRID_CELL_SIZE_REFLECTIVE" in a Landsat-8 MTL file. This external RS knowledge is then integrated with the initial image caption into a lengthy external knowledge passage. Subsequently, we use a defined specific prompt template for few-shot in-context learning to guide the LLM to condense relevant knowledge from four perspectives of human-like cognition and obtain multiview triplets. As shown in Table I, the prompt template we designed consists of four components: task instructions, lengthy external RS knowledge passages, triplet extraction instructions from four views, and a few demonstration examples. Among them, the task instructions form the publicly defined portion of the prompt template, outlining the condensation of relevant knowledge from four views, the OV, the AV, the SPV, and the SRV. The lengthy external knowledge passage consists of image caption and external RS knowledge. The triplet extraction instructions are derived from the instruction list we constructed. The examples are analogous to the triplet extraction task in terms of the human-like cognitive view, the extraction instructions, and the user intent. The triplet extraction instructions are derived from the instruction list we constructed, as shown in Table II. Specifically, the triplet extraction instructions from the OV are designed to condense knowledge revealing the inherent nature and category types of geographic entities. The triplet extraction instructions from the AV aim to condense knowledge associated with the specific properties of these entities. The triplet extraction instructions from the SPV are intended to condense knowledge reflecting the spatial locations of geographic entities. Moreover, the triplet extraction instructions from the SRV condense knowledge about the relationships between geographic entities in spatial distributions and spatial orientations. Instructions from the same view share the same natural language meaning. The examples are of the same task for LLM to learn the pattern in the form of demonstration. We create several demonstration examples artificially beforehand and employ a heuristic approach to select specific examples used in prompt templates. In this approach, the similarity between the example samples and the test samples is considered, as well as the diversity of different examples. Based on the results output from the LLM, the content of the prompt templates and the instruction and demonstration example selections are optimized and iterated until the model can output better results. As illustrated in Fig. 4, we take optical RS images containing the Taipei 101 building as an example. These images were captured by the WorldView-2 satellite on April 20, 2011, July 5, 2015, and May 22, 2018. The WorldView-2 satellite is equipped with both a panchromatic sensor and a multispectral sensor capable of providing RS images at a 0.5-m panchromatic resolution and a 1.8-m multispectral resolution, with a swath width of 16.4 km [106]. After image captions are generated via a generic VLM, external knowledge passages are built on the basis of RS external knowledge databases by extracting information corresponding to given RS images, such as names, shapes, areas, and geographic locations, as well as spatial relationships with the surroundings. Next, we adopt a designed prompt template to guide an LLM to condense relevant knowledge from four perspectives of human-like cognition and construct a view-level knowledge graph. By leveraging LLM with automated extraction capabilities, we significantly improve the efficiency and simplicity of the above knowledge distillation process, which lays a solid foundation for the subsequent representation of a view-level knowledge graph.

To ensure the efficacy of view-level knowledge representation, we map the triples in view-level knowledge graphs from different views into a low-dimensional vector space, vectorizing the structural and semantic information contained therein. First, we create four datasets to store triples in the view-level knowledge graph from the OV, AV, SPV, and SRV, denoted as $R = \{R_o, R_a, R_p, R_r\}$. Then, we employ a common

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TABLE I

PROMPT TEMPLATE FOR TRIPLET EXTRACTION FROM FOUR VIEWS

Prompt:

Extract triplets in the form of <Entity, Relation, Entity> from the following input text {Input text}. In addition, choose the most appropriate view to categorize every triplet. The available type of views are as follows: object view {Instruction from the view of object}, attribute view {Instruction from the view of attribute}, spatial position view {Instruction from the view of spatial position}, spatial relation view {Instruction from the view of spatial relation}. Where, the object view focuses primarily on the basic information of a specific object, including its name, type, and other identifiers. The attribute view emphasizes information that reveals the essential characteristics of an object, such as shape, color, area, function, width, and height. The spatial position view is concerned with the location and distribution characteristics of an object within a spatial context, including its geographic location and spatial distribution. Finally, the spatial relation view examines the spatial relationships between a particular object and other objects, which can be further divided into spatial orientation relationships, such as northwest, north, northeast, east, and southeast, and spatial topological relationships, including surrounding, intersecting, adjacent, and nearby.

Examples:

## Example 1	## Example 2	## Example 3	## Example 4
#Instruction: Extract triples in the format of (entity1, relation, entity2) focusing on the information of an object's inherent nature or type from the provided "Input Text" .	#Instruction: Using triplets in the format of (entity1, relation, entity2) to represent the spatial absolute position or distribution feature of an object in provided "Input text".	#Instruction: Extract triplets focusing solely on characteristics or properties of an object from provided "Input text" in the formatof (entity1, relation, entity2) .	#Instruction: Use triplets in the form of (entity1, relation, entity2) to express the spatial association relationship between two objects in the provided "Input Text".
INPUT: Xinghai Park has a large circular lawn in the center surrounded by trees and building.	**INPUT**: The building in the image is the Taipei101, which is a land mark skyscraper located in Taipei.	**INPUT**: Taipei101 covers an area of 30,277 m^2 and shaped like a huge pagoda.	**INPUT**: Taipei101 is next to Xinyi Plaza to the east, Xinyi No.21 Park to the north.
OUTPUT: <park, name is, Xinghai Park>, <park, has, lawn>	**OUTPUT**: <Taipei101, located in, Taipei>	**OUTPUT**: <Taipei101, area, 30,277 m^2>, <Taipei101, shape, pagoda>	**OUTPUT**: <Xinyi Plaza, to the east of, Taipei101>, <Xinyi No.21 Park, to the north of, Taipei101>

LLM: {List for triples extracted from four views}.

The black text represents the public definition of the template, the green text represents the instruction for triplet extraction, the red text represents the input knowledge context, and the blue text represents the answer from LLM.

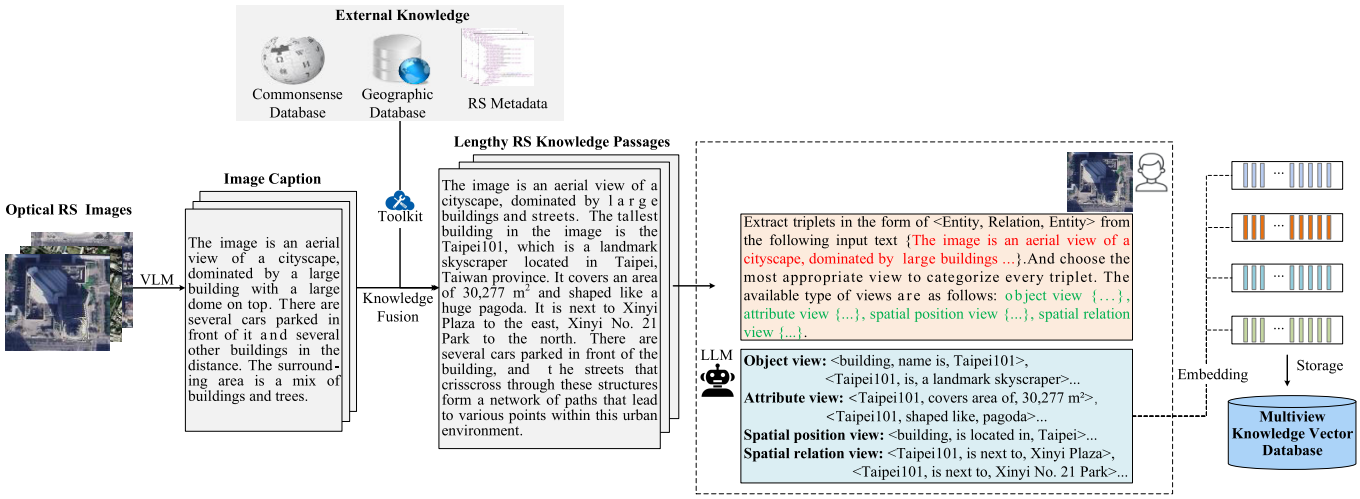


Fig. 4. Progress of view-based RSKG construction.

knowledge graph embedding model, TransH [95], to convert nodes (entities) and edges (relations) in the view-level knowledge graph into vectors. TransH is an improved method of TransE [96]. It allows each relation to be translated on a hyperplane, leading to better attribute capture performance, especially when handling one-to-many, many-to-one, and many-to-many relations. In practical implementation, we first initialize the entities and relations in each view dataset as vectors. The model is optimized by iteratively minimizing the distance between the sum of the head entity vector and relation vector and the tail entity vector in each triple to produce high-quality embedding vectors. Therefore, we obtain multiview knowledge vectors, including OV knowledge vectors, AV knowledge vectors, SPV knowledge vectors, and SRV knowledge vectors. Finally, we store these view knowledge vectors in the multiview knowledge vector database for subsequent contrastive learning of multigranularity RS semantic fusion.

B. Multigranularity RS Semantic Fusion

In multigranularity RS semantic fusion, we construct an asymmetric RS-M2CL to explore fine-grained semantics from multimodal inputs, including optical RS images, text, and view-level knowledge graphs, with the aim of achieving a comprehensive understanding of the underlying semantics. As shown in Fig. 5, RS-M2CL contains a visual encoder, a text encoder, a knowledge encoder, and a multimodal encoder, where the former three are used to encode the inputs of the three modalities to obtain the feature vectors. Then, a cross-attention mechanism is used to capture the highly correlated parts between different modalities. Finally, a multimodal encoder is employed to fuse multimodal features via the gated fusion method [97], [98], [99]. At this juncture, the precise extraction and representation of fine-grained semantics in optical RS images can be effectively achieved. For an RS image–text dataset $\boldsymbol{X} = \{X_1, X_2, \ldots, X_N\}$ with N samples, any sample X_i can be denoted as $X_i = (X_{\mathrm{I},i}, X_{\mathrm{L},i}, X_{\mathrm{K},i})$,

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TABLE II
INSTRUCTIONS FOR TRIPLET EXTRACTION FROM FOUR VIEWS

Object View
• Extract relational information that only focuses on the inherent nature or classification of an object.
• Provide triplets of this text which are only related to the inherent nature or classification of an object.
• Use triplets to express the information of an object's inherent nature or type.
• Present structured information focused on the inherent nature or classification of an object.
• Extract triplets focusing solely on the inherent nature or classification of an object from this text.
• Using triplets to represent the inherent nature or classification of an object in the input text.
Attribute View
• Extract relational information that only focuses on the characteristics or properties of an object.
• Provide triplets of this text which are only related to the characteristics or properties of an object.
• Use triplets to express an object's characteristics or properties.
• Present structured information focused on characteristics or properties of an object.
• Extract triplets focusing solely on characteristics or properties of an object from this text.
• Using triplets to represent characteristics or properties of an object in the input text.
Spatial Position View
• Extract relational information that only focuses on the spatial absolute position or distribution feature of an object.
• Provide triplets of this text which are only related to the spatial absolute position or distribution feature of an object.
• Use triplets to express the spatial absolute position or distribution feature of an object.
• Present structured information focused on the spatial absolute position or distribution feature of an object.
• Extract triplets focusing solely on the spatial absolute position or distribution feature of an object from this text.
• Using triplets to represent the spatial absolute position or distribution feature of an object in the input text.
Spatial Relation View
• Extract relational information that only focuses on the spatial association relationship between two objects.
• Provide triplets of this text which are only related to how entities are positioned or connected in space relative to each other.
• Use triplets to express how objects are positioned or connected in space relative to each other.
• Present structured information focused on the spatial association relationship between two objects.
• Extract triplets focusing solely on the spatial association relationship between two objects from this text.
• Using triplets to represent how entities are positioned or connected in space relative to each other in the input text.

where $X_{\mathrm{I},i}$ is an optical RS image, $X_{\mathrm{L},i}$ is a lengthy external knowledge passage corresponding to $X_{\mathrm{I},i}$, and $X_{\mathrm{K},i}$ is the multiview knowledge condensed from $X_{\mathrm{L},i}$ via the method in Section III-A. $X_{\mathrm{K},i}$ consists of OV knowledge $X^{\mathrm{O}}_{\mathrm{K},i}$, AV knowledge $X^{\mathrm{A}}_{\mathrm{K},i}$, SPV knowledge $X^{\mathrm{SP}}_{\mathrm{K},i}$, and SRV knowledge $X^{\mathrm{SR}}_{\mathrm{K},i}$.

1) Encoding of Different Modalities: We utilize the visual encoder in CLIP-ViT-B/32 for encoding the visual modality, extracting features from the optical RS image $X_{\mathrm{I},i}$ to obtain the visual feature vector $H_{\mathrm{I},i}$, denoted as

$$H_{\mathrm{I},i} = f_{\mathrm{IE}}\left(X_{\mathrm{I},i}\right) \tag{1}$$

where f_{IE} represents the processing of the visual encoder. Similarly, we use the text encoder in CLIP-ViT-B/32 for textual modality encoding, extracting features from the lengthy external knowledge passage $X_{\mathrm{L},i}$ to obtain textual feature vector $H_{\mathcal{L},i}$, denoted as

$$H_{\mathrm{L},i} = f_{\mathrm{LE}}\left(X_{\mathrm{L},i}\right) \tag{2}$$

where f_{LE} represents the processing of the text encoder. We extract the multiview knowledge feature vectors $H^{\mathrm{O}}_{\mathrm{K},i}$, $H^{\mathrm{A}}_{\mathrm{K},i}$, $H^{\mathrm{SP}}_{\mathrm{K},i}$, and $H^{\mathrm{SR}}_{\mathrm{K},i}$ from the knowledge vector database, which are obtained via the TransH method in Section III-A, and connect them as a view knowledge feature vector, $H_{\mathrm{K},i} = [H^{\mathrm{O}}_{\mathrm{K},i}; H^{\mathrm{A}}_{\mathrm{K},i}; H^{\mathrm{SP}}_{\mathrm{K},i}; H^{\mathrm{SR}}_{\mathrm{K},i}]$. [;] denotes the vector concatenation operation.

2) Contrastive Learning: To enhance the ability of RS-M2CL to learn high-level fine-grained semantics, we deploy contrastive learning to strengthen cross-granularity semantic alignment among features in different modalities. We define multigranularity contrastive learning (MGCL) loss as the learning object, which consists of interimage contrastive (IIC) loss, image–text contrastive (ITC) loss, and image-view knowledge contrastive (IVC) loss.

IIC loss is used to guide contrastive learning between optical RS images at different spatial scales within the image modality. It aims to integrate local, deep fine-grained semantics of optical RS images at large spatial scales with holistic, macroscopic coarse-grained semantics of optical RS images at small spatial scales. Therefore, we obtain consistent semantics of optical RS images at different spatial scales. Referring to [100], we define the IIC loss $\mathcal{L}_{\mathrm{IIC}}$ as

$$\mathcal{L}_{\mathrm{IIC}} = -\sum_{i=1}^{N} \log \frac{\exp\left(\mathrm{s}\left(H_{\mathrm{I},i}, H'_{\mathrm{I},i}\right)/\tau\right)}{\sum_{j=1}^{N} \exp\left(\mathrm{s}\left(H_{\mathrm{I},i}, H_{\mathrm{I},j}\right)/\tau\right)} \quad \text{st. } i \neq j \tag{3}$$

where $H_{\mathrm{I},i}$, $H'_{\mathrm{I},i}$, and $H_{\mathrm{I},j}$ are the visual feature vectors of anchor $X_{\mathrm{I},i}$, positive sample $X'_{\mathrm{I},i}$, and negative sample $X_{\mathrm{I},j}$, respectively. $X'_{\mathrm{I},i}$ is a new optical RS image obtained via image enhancement on $X_{\mathrm{I},i}$. $X_{\mathrm{I},j}$ is an optical RS image in $\boldsymbol{X}$ excluding $X_{\mathrm{I},i}$. $\mathrm{s}(\cdot,\cdot)$ denotes the cosine similarity between two feature vectors, and τ is the temperature parameter.

ITC loss is used to guide contrastive learning between optical RS images and lengthy external knowledge passages. It aims to complete cross-modal semantic alignment between the image and text modalities, enabling semantic complementarity and information enhancement. Referring to [100], we define the ITC loss $\mathcal{L}_{\mathrm{ITC}}$ as

$$\mathcal{L}_{\mathrm{ITC}} = -\sum_{i=1}^{N} \log \frac{\exp\left(\mathrm{s}\left(H_{\mathrm{I},i}, H_{\mathrm{L},i}\right)/\tau\right)}{\sum_{j=1}^{N} \exp\left(\mathrm{s}\left(H_{\mathrm{I},i}, H_{\mathrm{L},j}\right)/\tau\right)} \quad \text{st. } i \neq j \tag{4}$$

where $H_{\mathrm{I},i}$ is the visual feature vector of an anchor $X_{\mathrm{I},i}$, and $H_{\mathrm{L},i}$ and $H_{\mathrm{L},j}$ are the textual feature vectors of a positive sample $X_{\mathrm{L},i}$ and a negative sample $X_{\mathrm{L},j}$, respectively. $X_{\mathrm{L},i}$ is the lengthy external knowledge passage corresponding to $X_{\mathrm{I},i}$, and $X_{\mathrm{L},j}$ is a lengthy external knowledge passage in $\boldsymbol{X}$ excluding $X_{\mathrm{L},i}$.

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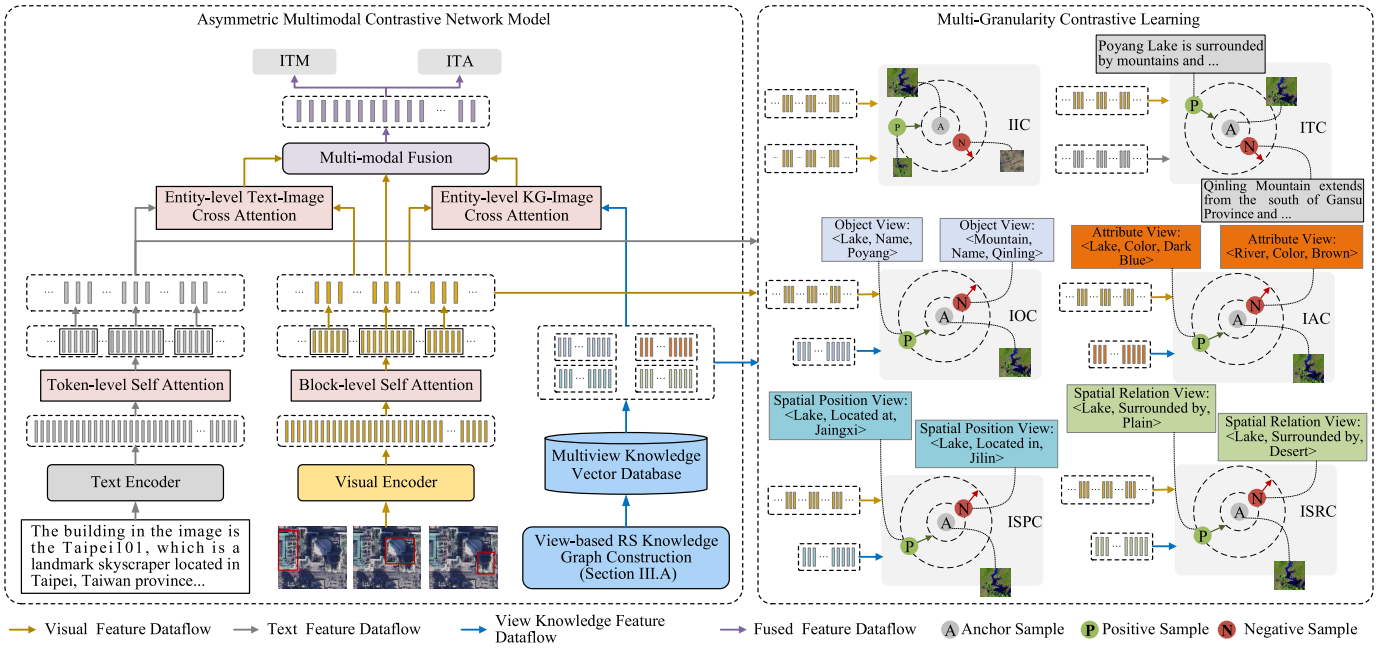


Fig. 5. Illustration of our proposed RS-M2CL.

IVC loss is used to guide contrastive learning between optical RS images and view-level knowledge graphs, thereby enhancing the completeness, accuracy, and semantic consistency of the understanding of optical RS images. IVC loss can be further expressed as a weighted combination of image-OV contrastive (IOC) loss, image-AV contrastive (IAC) loss, image-SPV contrastive (ISPC) loss, and image-SRV contrastive (ISRC) loss as

$$\begin{aligned} &\mathcal{L}_{\mathrm{IVC}} = \lambda_{\mathrm{O}}\mathcal{L}^{\mathrm{O}}_{\mathrm{IVC}} + \lambda_{\mathrm{A}}\mathcal{L}^{\mathrm{A}}_{\mathrm{IVC}} + \lambda_{\mathrm{SP}}\mathcal{L}^{\mathrm{SP}}_{\mathrm{IVC}} + \lambda_{\mathrm{SR}}\mathrm{L}^{\mathrm{SR}}_{\mathrm{IVC}} \\ &\text{st. } i \neq j, \lambda_{\mathrm{O}} + \lambda_{\mathrm{A}} + \lambda_{\mathrm{SP}} + \lambda_{\mathrm{SR}} = 1 \end{aligned} \tag{5}$$

where $\mathcal{L}^{\mathrm{O}}_{\mathrm{IVC}}$, $\mathcal{L}^{\mathrm{A}}_{\mathrm{IVC}}$, $\mathcal{L}^{\mathrm{SP}}_{\mathrm{IVC}}$, and $\mathcal{L}^{\mathrm{SR}}_{\mathrm{IVC}}$ denote the contrastive losses between the optical RS image and the OV knowledge, the AV knowledge, the SPV knowledge, and the spatial relation knowledge, respectively. $\lambda_{\mathrm{O}}, \lambda_{\mathrm{A}}, \lambda_{\mathrm{SP}}$, and λ_{SR} are used to balance their efforts in IVC loss. Given an example, the IOC loss $\mathcal{L}^{\mathrm{O}}_{\mathrm{IVC}}$ can be expressed as

$$\mathcal{L}^{\mathrm{O}}_{\mathrm{IVC}} = -\sum_{i=1}^{N} \log \frac{\exp\left(\mathrm{s}_{\mathrm{O}}\left(H_{\mathrm{I},i}, H^{\mathrm{O}}_{\mathrm{K},i}\right)/\tau\right)}{\sum_{j=1}^{N} \exp\left(\mathrm{s}_{\mathrm{O}}\left(H_{\mathrm{I},i}, H^{\mathrm{O}}_{\mathrm{K},j}\right)/\tau\right)} \tag{6}$$

where $H_{\mathrm{I},i}$ is the visual feature vector of an anchor $X_{\mathrm{I},i}$, and $H^{\mathrm{O}}_{\mathrm{K},i}$ and $H^{\mathrm{O}}_{\mathrm{K},j}$ represent the feature vectors of OV knowledge for positive sample $X^{\mathrm{O}}_{\mathrm{K},i}$ and negative sample $X^{\mathrm{O}}_{\mathrm{K},j}$, respectively. $X^{\mathrm{O}}_{\mathrm{K},i}$ is the OV knowledge corresponding to $X_{\mathrm{I},i}$, whereas $X^{\mathrm{O}}_{\mathrm{K},j}$ is the OV knowledge corresponding to any other optical RS image $X_{\mathrm{I},j}$ in $\boldsymbol{X}$ excluding $X_{\mathrm{I},i}$. $\mathrm{s}_{\mathrm{O}}(\cdot,\cdot)$ denotes the cosine similarity between two knowledge feature vectors in the OV.

Importantly, there is an internal relationship between the ISRC loss and the ISPC loss. The spatial relation similarity $\mathrm{s}_{\mathrm{SR}}(\cdot,\cdot)$ in the ISRC loss is a component of the spatial position similarity $\mathrm{s}_{\mathrm{SP}}(\cdot,\cdot)$ in the ISPC loss, which can be represented as

$$\mathrm{s}_{\mathrm{SP}}\left(H_{\mathrm{I},i}, H^{\mathrm{SP}}_{\mathrm{K},i}\right) = \omega_{\mathrm{DIS}}\mathrm{s}_{\mathrm{DIS}}\left(H_{\mathrm{I},i}, H^{\mathrm{SP}}_{\mathrm{K},i}\right) + \omega_{\mathrm{SR}}\mathrm{s}_{\mathrm{SR}}\left(H_{\mathrm{I},i}, H^{\mathrm{SR}}_{\mathrm{K},i}\right) \tag{7}$$

where $\mathrm{s}_{\mathrm{SP}}(\cdot,\cdot)$, $\mathrm{s}_{\mathrm{DIS}}(\cdot,\cdot)$, and $\mathrm{s}_{\mathrm{SR}}(\cdot,\cdot)$ denote the spatial position, spatial distance, and spatial relation similarity, respectively. $\mathrm{s}_{\mathrm{DIS}}(\cdot,\cdot)$ and $\mathrm{s}_{\mathrm{SR}}(\cdot,\cdot)$ can be represented by the distance between feature vectors, and ω_{DIS} and ω_{SR} are used to balance their efforts.

In general, MGCL loss $\mathcal{L}_{\mathrm{MGCL}}$ can be expressed as

$$\begin{aligned} &\mathcal{L}_{\mathrm{MGCL}} = \lambda_{\mathrm{IIC}}\mathrm{L}_{\mathrm{IIC}} + \lambda_{\mathrm{ITC}}\mathcal{L}_{\mathrm{ITC}} + \lambda_{\mathrm{IVC}}\mathcal{L}_{\mathrm{IVC}} \\ &\text{st. } \lambda_{\mathrm{IIC}} + \lambda_{\mathrm{ITC}} + \lambda_{\mathrm{IVC}} = 1 \end{aligned} \tag{8}$$

where λ_{IIC}, λ_{ITC}, and λ_{IVC} are used to balance the impacts of $\mathcal{L}_{\mathrm{IIC}}$, $\mathcal{L}_{\mathrm{ITC}}$, and $\mathcal{L}_{\mathrm{IVC}}$.

3) Modal Interaction: We use an attention module to achieve modal interaction, capturing content with high correlation across different modalities. As shown in Fig. 5, we design a two-level attention mechanism containing self-attention and cross-attention. Self-attention focuses on the high correlation between different parts within a single modality, with block-level self-attention (BSA) and token-level self-attention (TSA) applied to optical RS images and lengthy external knowledge passages, respectively.

BSA consists of M self-attention layers connected in sequence. The input of BSA is the block-level visual features of optical RS images, whereas the output is the entity-level visual features. There is a correlation between block-level visual features within the same entity; thus, these features are filtered and pooled in BSA for aggregation to obtain the entity-level visual features of optical RS images. TSA has a similar structure to BSA, in which the input is the token-level textual features of lengthy external knowledge passages. In TSA, token-level textual features with high relevance

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and belonging to the same entity spans are filtered and pooled to obtain the entity-level textual features of lengthy external knowledge passages.

Entity-level text–image cross-attention (ETICA) is mainly used to capture the correlations between optical RS images and lengthy external knowledge passages to accomplish entity-level modal interactions at the macrolevel. The entity that fully represents physical entities in the text and image modalities corresponds to a text sequence and an image block, respectively. ETICA has a multihead cross-attention structure consisting of multiple attention modules connected in parallel. Each module consists of M attention layers connected in sequence. In ETICA, we take the textual feature H_{L} as the query Q_{L} and the visual feature H_{I} as the key K_{I} and the value V_{I}. The text–image attentional feature $H_{\mathrm{L,I}}^{\mathrm{Attn}}$ can be derived through a cross-attention modal interaction process, where the relevant visual feature H_{I} from the optical RS image, aligned with the lengthy external knowledge passage, is identified and assigned higher attention

$$H_{\mathrm{L,I}}^{\mathrm{Attn}} = \mathrm{softmax}\left(\frac{Q_{\mathrm{L}} K_{\mathrm{I}}^{\mathrm{T}}}{\sqrt{d}_{K_{\mathrm{I}}}}\right) V_{\mathrm{I}} \tag{9}$$

where $d_{K_{\mathrm{I}}}$ is the dimension of K_{I}.

Entity-level KG-image cross-attention (EKICA) is mainly used to capture correlations between optical RS images and multiview knowledge to complete entity-level modal interactions at a microlevel across different views. EKICA consists of multiple attention modules connected in parallel, and different attention modules are used for the modal interaction between knowledge from different views and optical RS images. Each attention module consists of M attentions connected in sequence. In EKICA, we take the view knowledge feature H_{K} as the query Q_{K} and take the visual features of optical RS images H_{I} as the key K_{I} and the value V_{I} simultaneously. The KG-image attention feature $H_{\mathrm{K,I}}^{\mathrm{Attn}}$ is derived through a cross-attention modal interaction process, expressed as

$$H_{\mathrm{K,I}}^{\mathrm{Attn}} = \mathrm{softmax}\left(\frac{Q_{\mathrm{K}} K_{\mathrm{I}}^{\mathrm{T}}}{\sqrt{d}_{K_{\mathrm{I}}}}\right) V_{\mathrm{I}} \tag{10}$$

where $d_{K_{\mathrm{I}}}$ is the dimension of K_{I} [101].

4) Modal Fusion: Given that visual features, textual features, and multiview knowledge features contribute differently to the results of multigranularity RS semantic fusion, we adopt the gated fusion method [97], [98], [99] to fuse the features from these three distinct modalities, shield the differences in feature space distributions across modalities, and yield the final hybrid features for the multigranularity RS semantic fusion process. Referring to [121], the fusion vector H_{F} is defined as

$$H_{\mathrm{F}} = \left(1 - \lambda_{\mathrm{L,I}} - \lambda_{\mathrm{K,I}}\right) H_{\mathrm{I}} + \lambda_{\mathrm{L,I}} H_{\mathrm{L,I}}^{\mathrm{Attn}} + \lambda_{\mathrm{K,I}} H_{\mathrm{K,I}}^{\mathrm{Attn}} \tag{11}$$

where $\lambda_{\mathrm{L,I}}$ and $\lambda_{\mathrm{K,I}}$ are the two weight factors used to balance the efforts of H_{I}, $H_{\mathrm{L,I}}^{\mathrm{Attn}}$, and $H_{\mathrm{K,I}}^{\mathrm{Attn}}$. $\lambda_{\mathrm{L,I}}$ is defined as

$$\lambda_{\mathrm{L,I}} = \sigma\left(W_{\mathrm{I}}^{\mathrm{l}} H_{\mathrm{I}} + W_{\mathrm{L,I}}^{\mathrm{l}} H_{\mathrm{L,I}}^{\mathrm{Attn}} + W_{\mathrm{K,I}}^{\mathrm{l}} H_{\mathrm{K,I}}^{\mathrm{Attn}}\right) \tag{12}$$

where $\sigma(\cdot)$ denotes the gating function, and $W_{\mathrm{L}}^{\mathrm{l}}$, $W_{\mathrm{L,I}}^{\mathrm{l}}$, and $W_{\mathrm{K,I}}^{\mathrm{l}}$ are the learning parameters. $\lambda_{\mathrm{K,I}}$ is similar to $\lambda_{\mathrm{L,I}}$.

To facilitate more efficient training of the multimodal encoder and to manifest the respective strengths of textual, visual, and view knowledge features during modal fusion, we define multimodal fusion (MFS) learning (MMFL), which consists of image–text matching (ITM) loss and image–text alignment (ITA) loss.

Referring to [30], we design an ITM loss to predict the matchability among image, text, and view knowledge, expressed as

$$\mathcal{L}_{\mathrm{ITM}} = -\sum_{i,j,k=1}^{N} y_{i,j,k} \log\left(p\left(H_{\mathrm{I},i}, H_{\mathrm{L,I},j}^{\mathrm{Attn}}, H_{\mathrm{K,I},k}^{\mathrm{Attn}}\right)\right) \tag{13}$$

where $y_{i,j,k} \in \{\pm 1\}$ denotes the ground-truth label for the matchability of the optical RS image, text, and multiview knowledge. The value assigned to $y_{i,j,k}$ is 1 when all three are matched and -1 otherwise. $p(\cdot,\cdot,\cdot) \in [0, 1]$ denotes the degree of predicted matchability among these three features.

ITA guides the learning process of the multimodal encoder by comparing the difference in matchability between the original optical RS image with external knowledge passages and multiview knowledge and matchability between the entity-masked optical RS image with external knowledge passages and multiview knowledge. The former tends to exhibit a higher matchability score, thereby emphasizing the absence of entities in the optical RS image. We define the ITA loss $\mathcal{L}_{\mathrm{ITA}}$ as

$$\mathcal{L}_{\mathrm{ITA}} = \sum_{i}^{N} \max\Big(0, -y_{i,j,k}\big(p\left(H_{\mathrm{I},i}, H_{\mathrm{L,I},j}^{\mathrm{Attn}}, H_{\mathrm{K,I},k}^{\mathrm{Attn}}\right) - p_{\mathrm{m}}\left(H_{\mathrm{I},i}, H_{\mathrm{L,I},j}^{\mathrm{Attn}}, H_{\mathrm{K,I},k}^{\mathrm{Attn}}\right)\big)\Big) \tag{14}$$

where $p_{\mathrm{m}}(\cdot,\cdot,\cdot) \in [0, 1]$ denotes the matchability score in terms of the optical RS image with a masked entity.

Therefore, the MMFL loss $\mathcal{L}_{\mathrm{MMFL}}$ can be expressed as

$$\mathcal{L}_{\mathrm{MMFL}} = \lambda_{\mathrm{ITM}} \mathcal{L}_{\mathrm{ITM}} + \lambda_{\mathrm{ITA}} \mathcal{L}_{\mathrm{ITA}}. \tag{15}$$

Here, λ_{ITM} and λ_{ITA} are used to balance the contributions of the ITM and ITA to the MMFL loss.

C. Data Bootstrapping

RS image–text samples with rich semantics and high quality directly impact the performance and training efficiency of the RSMVKF. To address the limitations caused by the characteristics of optical RS images, we devise an optical RS image enhancement method. We propose three different sampling methods for the multigranularity contrastive learning process described in Section III-B: multiscale image sampling, multimodal image–text sampling, and knowledge-based multiview image–text sampling.

1) Optical RS Image Enhancement: Although there has been a notable increase in the acquisition capability of optical RS images, the available training datasets remain limited due to their inherent characteristics. On the one hand, multitemporal and multiscale optical RS images exhibit significant differences, attributed to their intraclass heterogeneity and interclass homogeneity. On the other hand, the quality of optical RS images is usually affected by external factors, such as cloud coverage, atmospheric conditions, and illumination variations. To mitigate model overfitting arising from

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small-scale training datasets, enhancing optical RS images to diversify training data while maintaining semantic consistency with the original images is crucial. Therefore, considering the aforementioned characteristics of optical RS images in terms of optical imaging methods, spatial and temporal natures, and external factors, we conduct RS image enhancement through the following three steps.

1) In light of the multitemporal nature of optical RS images, there are feature dissimilarities between images captured at different times from the same location. We construct a sequence of multitemporal RS images with the same spatial range and spatial scale, which are taken at different times.
2) In view of the multiscale nature of optical RS images, which exhibit obvious feature differences in the same physical entity at different spatial scales, we resample RS images to obtain new images with different spatial scales and then perform the spatial pyramid technique to crop them sequentially into tiles, constructing a multiscale optical RS image sequence.
3) Considering the heightened susceptibility of optical RS images to external factors such as light, cloud coverage, and atmospheric conditions, which complicate the process of image feature extraction, we adopt the random local erasure technique to simulate these characteristics. By randomly placing opaque regions of varying sizes in different areas of each RS image, we eliminate local image information, thereby enhancing the robustness of the model.

2) Multiscale Image Sampling: Multiscale image sampling involves creating samples from optical RS images captured at different spatial scales, aiming at integrating fine-grained features at large spatial scales with coarse-grained features at small spatial scales. For a minibatch $X_{(N)}$, optical RS image enhancement is performed on an RS image $X_{\mathrm{I},i}$ to obtain a new image with a different spatial scale $X'_{\mathrm{I},i}$. As shown in Fig. 6(a), for any optical RS image $X_{\mathrm{I},i}$ taken as an anchor, its corresponding enhanced image $X'_{\mathrm{I},i}$ with a different spatial scale is taken as a positive sample, forming a positive pair $\{(X_{\mathrm{I},i}, X'_{\mathrm{I},i})\}$; any optical RS image $X_{\mathrm{I},j}$ in $X_{(N)}$ except $X_{\mathrm{I},i}$ is taken as a negative sample, forming $N-1$ negative pairs $\{(X_{\mathrm{I},i}, X_{\mathrm{I},j})|j \in \{1,2,\ldots,N\}\backslash i\}$.

3) Multimodal Image–Text Sampling: Multimodal image–text sampling involves constructing samples consisting of optical RS images and corresponding view-level knowledge texts to achieve the fusion of visual and textual features via intermodal semantic complementarity. As shown in Fig. 6(b), for a minibatch $X_{(N)}$, we use optical RS images and lengthy external knowledge passages to construct multimodal sample pairs. For any optical RS image $X_{\mathrm{I},i}$ in $X_{(N)}$ taken as an anchor, the corresponding lengthy external knowledge passages $X_{\mathrm{L},i}$ related to $X_{\mathrm{I},i}$ are taken as a positive sample, forming a positive pair $\{(X_{\mathrm{I},i}, X_{\mathrm{L},i})\}$; we take other lengthy external knowledge passages $X_{\mathrm{L},j}$ in $X_{(N)}$ excluding $X_{\mathrm{L},i}$ as negative samples to construct $N-1$ negative pairs $\{(X_{\mathrm{I},i}, X_{\mathrm{L},j})|j \in \{1,2,\ldots,N\}\backslash i\}$.

4) Knowledge-Based Multiview Image–Text Sampling: Knowledge-based multiview image–text sampling constructs samples that incorporate optical RS images and multiview knowledge. It aims to enhance the semantic consistency in the understanding of optical RS images by using knowledge from different views. Considering contrastive learning on different views, knowledge-based multiview image–text sampling encompasses OV image–text sampling, AV image–text sampling, SPV image–text sampling, and SRV image–text sampling.

1) *OV Image–Text Sampling:* As shown in Fig. 6(c), for an optical RS image $X_{\mathrm{I},i}$ in the minibatch $X_{(N)}$ taken as the anchor, we take the OV knowledge $X^{\mathrm{O}}_{\mathrm{K},i}$ corresponding to $X_{\mathrm{I},i}$ as a positive sample to obtain an OV positive pair $\{(X_{\mathrm{I},i}, X^{\mathrm{O}}_{\mathrm{K},i})\}$ and take the OV knowledge other than $X^{\mathrm{O}}_{\mathrm{K},i}$ as negative samples to construct $(N-1)$ OV negative pairs $\{(X_{\mathrm{I},i}, X^{\mathrm{O}}_{\mathrm{K},j})|j \in \{1,2,\ldots,N\}/i$.
2) *AV Image–Text Sampling:* As shown in Fig. 6(d), for an optical RS image $X_{\mathrm{I},i}$ in the minibatch $X_{(N)}$ taken as the anchor, we take the AV knowledge $X^{\mathrm{A}}_{\mathrm{K},i}$ corresponding to $X_{\mathrm{I},i}$ as a positive sample to construct an AV positive pair $\{(X_{\mathrm{I},i}, X^{\mathrm{A}}_{\mathrm{K},i})\}$ and take any other AV knowledge in $X_{(N)}$ except $X^{\mathrm{A}}_{\mathrm{K},i}$ as the negative sample to construct $(N-1)$ AV negative pairs $\{(X_{\mathrm{I},i}, X^{\mathrm{A}}_{\mathrm{K},j})|j \in \{1,2,\ldots,N\}/i$.
3) *SPV Image–Text Sampling:* As shown in Fig. 6(e), for an optical RS image $X_{\mathrm{I},i}$ in the minibatch $X_{(N)}$ taken as the anchor, we take the SPV knowledge $X^{\mathrm{SP}}_{\mathrm{K},i}$ corresponding to $X_{\mathrm{I},i}$ as the positive sample and take any SPV knowledge $X^{\mathrm{SP}}_{\mathrm{K},j}$ in $X_{(N)}$ as the negative sample. Then, we obtain an SPV positive pair $\{(X_{\mathrm{I},i}, X^{\mathrm{SP}}_{\mathrm{K},i})\}$ and $(N-1)$ SPV negative pairs $\{(X_{\mathrm{I},i}, X^{\mathrm{SP}}_{\mathrm{K},j})|j \in \{1,2,\ldots,N\}/i$.
4) *SRV Image–Text Sampling:* As shown in Fig. 6(f), for an optical RS image $X_{\mathrm{I},i}$ in the minibatch $X_{(N)}$ taken as the anchor, we take an SRV knowledge $X^{\mathrm{SR}}_{\mathrm{K},i}$ corresponding to $X_{\mathrm{I},i}$ as the positive sample and take any SRV knowledge $X^{\mathrm{SR}}_{\mathrm{K},j}$ in $X_{(N)}$ as the negative sample. Then, we obtain an SRV positive pair $\{(X_{\mathrm{I},i}, X^{\mathrm{SR}}_{\mathrm{K},i})\}$ and $(N-1)$ SRV negative pairs $\{(X_{\mathrm{I},i}, X^{\mathrm{SR}}_{\mathrm{K},j})|j \in \{1,2,\ldots,N\}/i$.

IV. Experimental Evaluation and Analysis

To verify the ability of the proposed method (RSMVKF) to understand optical RS images across different applications, we divide our experimental tasks into two categories: perception tasks (Sections IV-C2–IV-C4) and reasoning tasks (Sections IV-C5–IV-C7), allowing for a comparative analysis with other advanced methods. The purpose of the perception tasks is to extract information and features from images, enabling the discovery and recognition of objects, scenes, and structures within RS images. These tasks include RSSC, RSOD, and RS semantic retrieval (RSSR). In contrast, the reasoning tasks focus on a deeper understanding and semantic analysis of image content, enabling the model to answer questions related to the image. These tasks include RSCP, RSVG, and RSVQA.

A. Dataset Description

1) View-Based Knowledge Graph Extraction Dataset: We present an image–text paired dataset with lengthy external

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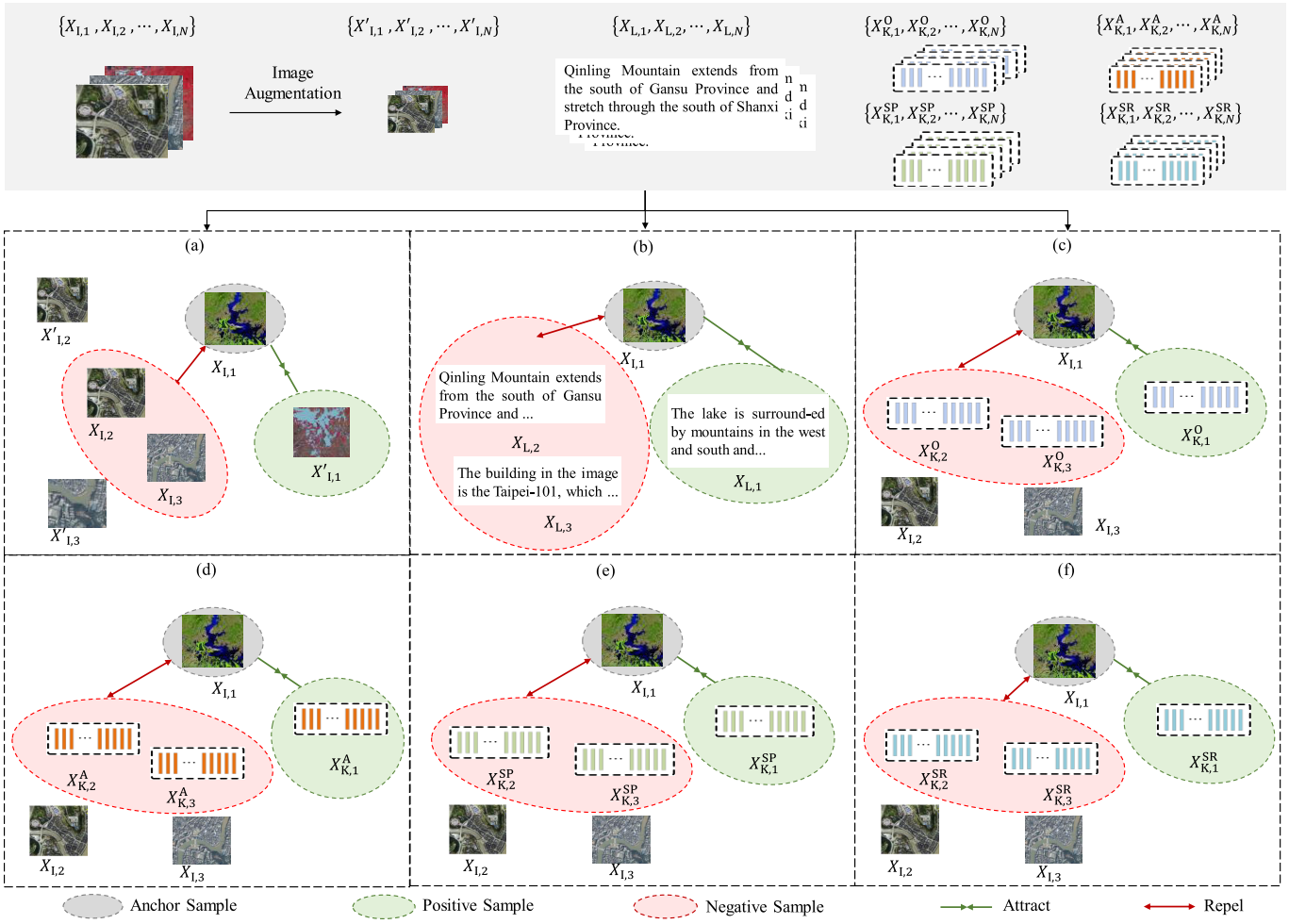


Fig. 6. Illustration of (a) multiscale image sampling, (b) multimodal image–text sampling, (c) knowledge-based multiview image–text sampling, (d) knowledge-based multiview image–text sampling (attribute view), (e) knowledge-based multiview image–text sampling (spatial position view), and (f) knowledge-based multiview image–text sampling (SRV).

knowledge passages in the field of optical RS, RSVKG100H. Specifically, we select geographically diverse regions and various types of objects, collecting 1300 sample optical RS images with different spatial scales taken at different times from three platforms, such as TerraColor NextGen Imagery [102], Google Earth [103], and Esri World Imagery [104], as detailed in Table III. These optical RS images are extracted from various satellite products, including Landsat 8 [105], WorldView-2/-3 [106], [107], GeoEye-1 [108], and SPOT6 PHR1A [109], covering 160 physical entities distributed in 18 spatial regions. The corresponding external knowledge texts are composed of a series of generative captions of optical RS images and external knowledge databases, yielding 5087 samples. The view knowledge is further condensed on the basis of the lengthy external knowledge passages, forming a dataset containing 14 566 view triplets.

2) Optical RS Image Downstream Task Datasets: To ensure the validity of the evaluation results, we select typical experimental datasets corresponding to each downstream task for utilization in optical RS image downstream perception and reasoning tasks. These datasets are built via optical RS images from products such as WorldView-2/-3, GeoEye-1, Sentinel-2, and NAIP, which are accessible on platforms such as Google Earth and Tianditu. The satellite information, resolution, the number of samples, and other relevant details are provided in Table IV.

B. Implementation Details

In the view-based knowledge graph construction stage, we employ MiniGPT-4 [17] as the VLM to generate image captions, whereas GPT-4o-2024-08-06 [119] serves as the LLM to condense view triples from the lengthy external knowledge passages, which is formed by fusing the image captions with external knowledge. We subsequently take several measures to address the issues of incorrectness and incompleteness that arise from the direct employment of pretrained LLMs in downstream tasks. We conduct few-shot in-context learning on the LLM, insert processing modules between the output of the LLM and the subsequent knowledge encoder, and adjust the temperature parameter during model inference. In the multigranularity RS semantic fusion stage, we subsequently use the pretrained CLIP-ViT-B/32 to build the backbone structure of RS-M2CL. The visual encoder and text encoder of CLIP-ViT-B/32 are utilized as the visual encoder and the text encoder, respectively, in RS-M2CL,

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TABLE III
SUMMARY OF THE DATASET USED FOR VIEW-BASED KNOWLEDGE CONDENSATION

Modality	Image Source	Satellite	Sensor	Resolution	Time Range	Samples
Optical RS Image	TerraColor NextGen Imagery [102]	Landsat8[105]	OLI, TIRS	30 m	2014-2020	0.15k
	Google Earth [103]	Landsat8	MS	2 m	2015-2023	0.8k
		WorldView-2[106]	MS	1.2 m	2015-2023	
		WorldView-3[107]	PAN, MS	1.64 m	2015-2023	
		GeoEye-1[108]	PAN, MS	1.5 m	2018-2023	
		SPOT6[109]	PAN, MS	0.5 m	2018-2023	
	Esri World Imagery [104]	WorldView-2	MS	2 m	2014-2022	0.35k
		WorldView-3	MS	1.2 m	2015-2023	
		GeoEye-01	PAN, MS	1.64 m	2021	
Modality	Knowledge Source	Knowledge Type				Samples
External RS Knowledge Text	Common-Sense Knowledge	Wikipedia ConceptNet				4.8k
	Basic Geographic Information	GeoNames OpenStreetMap				0.18k
	Optical RS Data Metadata File	Landsat8 Metadata WorldView-2 Metadata WorldView-2 Metadata GeoEye01 Metadata SPOT6 Metadata PHR1A Metadata				0.1k

TABLE IV
SUMMARY OF THE DATASETS USED FOR DIFFERENT TASKS

Task	DataSet	Satellite/Product	Resolution	Samples	Task	DataSet	Satellite/Product	Resolution	Samples
RS Scene Classification (RSSC)	AID [110]	WorldView-2	2 m	10k	RS Image Captioning (RSCP)	RSICD [113]	WorldView-2	2 m	18.2k
	UCM [111]	NAIP	1 m	2.1k			WorldView-3	1.2 m	
RS Object Detection (RSOD)	NWPU VHR-10 [112]	WorldView-2 WorldView-3 PHR1A	2 m 1.2 m 0.5 m	0.8k			GeoEye1 PHR1A	1.64 m 0.5 m	
						UCM-Caption[115]	NAIP	1 m	10.5k
	DIOR [58]	WorldView-2 WorldView-3 GeoEye1 QuickBird	2 m 1.2 m 1.64 m 2.5 m	23.5k	RS Visual Grounding (RSVG)	RSVGD[116]	WorldView-2 WorldView-3 GeoEye1 PHR1A	2 m 1.2 m 1.64 m 0.5 m	4.2k
RS Semantic Retrieval (RSSR)	RSICD [113]	WorldView-2 WorldView-3 GeoEye1 PHR1A	2 m 1.2 m 1.64 m 0.5 m	18.2k		DIOR-RSVG[117]	WorldView-2 WorldView-3 GeoEye1 QuickBird	2 m 1.2 m 1.64 m 2.5 m	17.4k
	RSITMD [114]	WorldView-2 WorldView-3 GeoEye1 PHR1A	2 m 1.2 m 1.64 m 0.5 m	21.8k	RS Visual Question Answering (RSVQA)	RSVQA-LR [118]	Sentinel-2	10 m	77.2k
						RSVQA-HR [118]	NAIP	1 m	955.6k

whereas the TransH model is employed as the knowledge encoder. Trainable linear projection layers are appended to each encoder to align the feature dimensions. A gated fusion network is constructed using a linear projection layer and the sigmoid activation function to complete multimodal encoding. Consequently, we use a two-layer lightweight transformer decoder for generative tasks. The encoders and the decoder are kept frozen on the basis of their pretrained weights, and linear projection layers are randomly initialized for model training. This training process is conducted on a single NVIDIA Tesla A100 40G GPU, with a mini-batch size set to 128 and a learning rate of 1e−5. The parameter settings in the network are detailed in Table V. There are approximately 151M parameters in our RSMVKF and 170M parameters in the RSMVKF with a decoder. The parametric quantities of the comparison methods in different tasks are given in Table VI.

Moreover, we carry out an optimization on the model structure to alleviate the problem of diminished efficiency. Specifically, we employ DeepSpeed [120] to deploy and fine-tune RS-M2CL, utilizing training optimization strategies such as model compression containing pruning and parameter quantization techniques. We use the DeepSpeed platform to facilitate distributed training across multiple GPUs and nodes, and we set the learning rate t to 0.0001 and train the model for 300 epochs. The model training task is distributed across multiple GPUs through the utilization of model parallelism and data parallelism techniques, enabling the parallel processing of large models on multiple computing devices and leading to

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TABLE V
PARAMETERS SETTING OF THE RSMVKF

Parameters	Value
Temperature coefficient τ	0.08
IIC coefficient λ_{IIC}	0.1
ITC coefficient λ_{ITC}	0.35
IVC coefficient λ_{IVC}	0.55
ITA coefficient λ_{ITA}	0.25
ITM coefficient λ_{ITM}	0.75
IOC coefficient λ_{O}	0.4
IAC coefficient λ_{A}	0.3
ISPC coefficient λ_{SP}	0.15
ISRC coefficient λ_{SR}	0.15
Distance Similarity coefficient ω_{DIS}	0.7
Relation Similarity coefficient ω_{SR}	0.3
Minibatch m	128
Learning rate η	1e-5

TABLE VI
MODALITY AND PARAMETERS OF THE RSMVKF AND COMPARED METHODS

Task	Methods	Multimodal	Parameters
RS Scene Classification (RSSC)	EAM	×	85 M
	DFAGCN	×	138 M
	MA-FE	×	46 M
	MaPle	√	86 M
	LVM-StARS	√	86.75 M
	TeAw	√	86.75 M
RS Object Detection (RSOD)	PISLM	×	32 M
	SGPLM-IR	×	138 M
	MOL	×	36.46 M
	TSF-RGR	√	20.91 M
	TEMO	√	45 M
	MSAM	√	87 M
RS Semantic Retrieval (RSSR)	FBCLM	√	149 M
	MLT	√	149 M
	RemoteCLIP	√	304 M
RS Image Captioning (RSCP)	RSGPT	√	7 B
	RS-CapRet	√	7 B
	SkyEyeGPT	√	10.5 B
RS Visual Grounding (RSVG)	MSAM	√	87 M
	MSCMA	√	166.3 M
	LPVA	√	87 M
RS Visual Question Answering (RSVQA)	MiniGPT-4	√	7 B
	MiniGPTv2	√	7 B
	Qwen-VL	√	7.7 B
	LLaVA-1.5	√	13 B
	RSGPT	√	7 B
	GeoChat	√	13 B
	SkyEyeGPT	√	7 B
RSMVKF (ours)		√	151 M
RSMVKF-decoder (ours)		√	170 M

improvement in training speed. In model pruning, we focus on the importance indicators of different parameters to be deleted to select parameters, including the absolute value of the size of the parameter or its contribution to the loss function, and the method of pruning only the unimportant connections in the network while trying to maintain the performance of the network. In parameter quantization, we convert parameters in RS-M2CL, including weights, biases, and activation values, from high-precision to low-precision 8-bit integers to reduce the computational resource requirements and speed up the inference process.

C. Performance Analysis

1) View-Based Knowledge Condensation: We compare the condensation results of view-based knowledge graphs via several currently popular LLMs. The models selected for the experiment include GPT-4o-2024-08-06, LLama-3.1-405b [121], Qwen1.5-110b [122], Gemini-1.5-pro-api-0514 [123], Nemotron-4-340b [124], Claude-3.5-sonnet-20240620 [125], and Yi-Large [126]. The experimental data are drawn from view-based knowledge condensation datasets, which include optical RS images with different spatial scales captured at various times across globally distributed geographic regions and featuring diverse types of objects.

First, we use MiniGPT-4 to generate image captions for these RS images and then construct lengthy external knowledge passages by integrating external knowledge. Using the constructed prompt template and instruction lists, we evaluate seven LLMs for their ability to extract view triples from these lengthy external knowledge passages. The results, presented in Fig. 7, illustrate the varying effectiveness of these LLMs in extracting multiview triples. Among the tested models, GPT-4o-2024-08-06 achieves the highest performance, accurately extracting the four view triples from the lengthy external knowledge passages and providing precise and succinct entity and relation information. LLama-3.1-405b achieves the second-best performance but has several shortcomings, such as failing to extract OV triples from the sixth external knowledge passage. The remaining models display various deficiencies in view of triple extraction. For example, Qwen1.5-110b incorrectly categorizes the museum information as an AV, Gemini-1.5-pro-api-0514 misclassifies the spatial relationship between the airport and the city as an AV, and Yi-Large performs the worst, with significant deviations from the task instructions. Additionally, the view triples extracted by Qwen1.5-110b, Gemini-1.5-pro-api-0514, and Claude-3.5-sonnet-20240620 contain redundant information. For example, in the fourth external knowledge passage, the OV triple extracted by Qwen1.5-110b includes the relation "is a landmark skyscraper" and the entity "located in Taipei," covering both the OV and the SPV. Gemini-1.5-pro-api-0514 redundantly extracts two view triples for the spatial relation between Taipei101 and Xinyi Plaza. Consequently, we select GPT-4o-2024-08-06 as the generative LLM for view-level knowledge graph construction.

2) RS Scene Classification: The objective of RSSC is to analyze the content of optical RS images and divide them into different scene categories. In our RSSC experiment, we concentrate on a concrete task, i.e., land-use scene classification, which is susceptible to the issue of confusability. We compare the performance of the RSMVKF with that of EAM [127], DFAGCN [128], MA-FE [129], MaPle [130], LVM-StARS [131], and TEAW [132] to assess the effectiveness of the RSMVKF in classifying RS images into distinct land-use scene categories, such as residential, industrial, commercial, and farmland.

As shown in Table VI, we chose representative subsets of optical RS images from the UCM dataset [111] and the AID dataset [110], which present a certain degree of confusability,

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Method	(1) 	(2) 	(3) 
	Knowledge context: The image shows a large sports complex, named the Shenzhen Universiade Sports Centre, located in Shenzhen, and covers a total area of 520,000 square meters. The complex features a unique triangular structure, comprising a main stadium, a gymnasium, and a swimming pool, all connected by the surface of Dayun Lake. Several buildings surround the sports complex, and the surrounding area is covered in trees and other vegetation.	**Knowledge context:** The image shows a large, modern building with a large, curved roof and a large, circular base. The building is Canton Tower, located in Guangzhou. It is adjacent to the Pearl River and there are several smaller buildings nearby. The main body of the Canton Tower is 454 meters, the antenna mast is 146 meters, and the total height is 600 meters. It is the tallest TV tower in China.	**Knowledge context:** The image shows a large park with a lake in the center. The park is the South Park of Beijing Olympic Forest Park and covers an area of 380 hectares. And it is surrounded by trees and has a few walking paths. The lake is surrounded by trees and has a few small irregular islands in the middle. The park is surrounded by a large road that leads to the city.
GPT-4o-2024-08-06	**Object View** <sports complex, is named, Shenzhen Universiade Sports Centre> <sports complex, comprises, main stadium, gymnasium, swimming pool> **Attribute View** <sports complex, covers, 520,000 square meters> <sports complex, features, unique triangular structure> **Spatial Position View** <sports complex, located in, Shenzhen> **Spatial Relation View** <main stadium, gymnasium, swimming pool, are connected by, surface of Dayun Lake> <buildings, surround, sports complex>	**Object View** <building, is, Canton Tower> <Canton Tower, is, tallest TV tower in China> **Attribute View** <building, has, large curved roof> <building, has, large circular base> <main body of Canton Tower, is, 454 meters> <antenna mast, is, 146 meters> <total height of Canton Tower, is, 600 meters> **Spatial Position View** <Canton Tower, located in, Guangzhou> **Spatial Relation View** <Canton Tower, is adjacent to, Pearl River> <smaller buildings, are nearby, Canton Tower>	**Object View** <park, is, South Park of Beijing Olympic Forest Park> **Attribute View** <park, covers, 380 hectares> <park, has, few walking paths> <lake, has, few small irregular islands in middle> **Spatial Position View** <lake, is in, center of park> **Spatial Relation View** <park, is surrounded by, trees> <lake, is surrounded by, trees> <park, is surrounded by, large road> <road, leads to, city>
Nemotron-4-340b	**Object View** <a large sports complex, named, Shenzhen Universiade Sports Centre> **Attribute View** <a large sports complex, covers, a total area of 520,000 square meters> <a large sports complex, features, a unique triangular structure> **Spatial Position View** <a large sports complex, located, Shenzhen> **Spatial Relation View** <a large sports complex, comprises, a main stadium> <a large sports complex, comprises, a gymnasium> <a large sports complex, comprises, a swimming pool> <a large sports complex, connected by, the surface of Dayun Lake> <Several buildings, surround, the complex>	**Object View** <Canton Tower, is a, large modern building> <Canton Tower, is, tallest TV tower in China> **Attribute View** <Canton Tower, has, large curved roof> <Canton Tower, has, large circular base> <antenna mast of Canton Tower, has height, 146 meters> <Canton Tower, has total height, 600 meters> **Spatial Position View** <Canton Tower, located in, Guangzhou> **Spatial Relation View** <Canton Tower, adjacent to, Pearl River> <several smaller buildings, near, Canton Tower>	**Object View** <The park, is, South Park of Beijing Olympic Forest Park> **Attribute View** <The park, covers, 380 hectares> <The park, has, a few walking paths> <The lake, has, a few small irregular islands> **Spatial Position View** <The lake, is in, the center of the park> **Spatial Relation View** <The park, is surrounded by, trees> <The lake, is surrounded by, trees> <The park, is surrounded by, a large road> <The large road, leads to, the city>
Claude-3.5-sonnet-20240620	**Object View** <Shenzhen Universiade Sports Centre, is a, sports complex> <Shenzhen Universiade Sports Centre, located in, Shenzhen> <Shenzhen Universiade Sports Centre, comprises, main stadium, gymnasium, swimming pool> **Attribute View** <Shenzhen Universiade Sports Centre, has size, large> <Shenzhen Universiade Sports Centre, covers area, 520,000 square meters> <Shenzhen Universiade Sports Centre, has structure, triangular> **Spatial Position View** <buildings, surround, Shenzhen Universiade Sports Centre> <trees and vegetation, cover, surrounding area> **Spatial Relation View** <main stadium, gymnasium, swimming pool, connected by, Dayun Lake>	**Object View** <Canton Tower, is a, building> <Canton Tower, is a, TV tower> <Canton Tower, located in, Guangzhou> **Attribute View** <Canton Tower, has roof shape, curved> <Canton Tower, has base shape, circular> <Canton Tower, has main body height, 454 meters> <Canton Tower, has antenna mast height, 146 meters> <Canton Tower, has total height, 600 meters> <Canton Tower, is, tallest TV tower in China> **Spatial Position View** <Canton Tower, adjacent to, Pearl River> **Spatial Relation View** <Canton Tower, surrounded by, smaller buildings>	**Object View** <South Park of Beijing Olympic Forest Park, is a, park> **Attribute View** <South Park of Beijing Olympic Forest Park, has area of, 380 hectares> **Spatial Position View** <lake, is located in, center> <small irregular islands, are located in, middle of lake> **Spatial Relation View** <South Park of Beijing Olympic Forest Park, contains, lake> <South Park of Beijing Olympic Forest Park, is surrounded by, trees> <South Park of Beijing Olympic Forest Park, has, walking paths> <lake, is surrounded by, trees> <lake, contains, small irregular islands> <South Park of Beijing Olympic Forest Park, is surrounded by, large road> <large road, leads to, city>
Yi-Large	**Object View** <Shenzhen Universiade Sports Centre, named, sports complex> **Attribute View** <Shenzhen Universiade Sports Centre, covers area, 520,000 square meters> <Shenzhen Universiade Sports Centre, structure, triangular> <surrounding area, covered in, trees and other vegetation> **Spatial Position View** <Shenzhen Universiade Sports Centre, located in, Shenzhen> **Spatial Relation View** <Shenzhen Universiade Sports Centre, comprises, gymnasium> <Shenzhen Universiade Sports Centre, comprises, swimming pool> <Shenzhen Universiade Sports Centre, connected by, surface of Dayun Lake> <sports complex, surrounded by, several buildings>	**Object View** <Canton Tower, is a, modern building> <Canton Tower, is, the tallest TV tower in China> **Attribute View** <Canton Tower, has, a large curved roof> <Canton Tower, has, a large circular base> <Canton Tower, height of main body, 454 meters> <Canton Tower, height of antenna mast, 146 meters> <Canton Tower, total height, 600 meters> **Spatial Position View** <Canton Tower, located in, Guangzhou> **Spatial Relation View** <Canton Tower, adjacent to, Pearl River>	**Object View** <South Park of Beijing Olympic Forest Park, is a, large park> <The lake, is a, lake in the center of the South Park of Beijing Olympic Forest Park> <South Park of Beijing Olympic Forest Park, has, a few walking paths> **Attribute View** <South Park of Beijing Olympic Forest Park, covers, 380 hectare> <The lake, has, a few small irregular islands in the middle> **Spatial Position View** <The park, is located in, Beijing> <The lake, is located in, the center of the park> **Spatial Relation View** <The park, surrounded by, a large road that leads to the city> <The lake, is surrounded by, trees> <The park, has, a few walking paths>

Fig. 7. Comparison of several LLMs on triplet extraction. The red text represents incorrect triplets, the blue text represents missing triplets, and the orange text represents triples with redundant information.

such as desert and bare land, to facilitate comparisons with current popular end-to-end RSSC methods. We take the average accuracy (AA) and overall accuracy (OA) as the evaluation metrics. AA is the mean value of the classification accuracy among different RS scene categories, and OA is the ratio of correctly classified samples to the total number of samples in the dataset. AA and OA can be expressed as

$$\mathrm{AA} = \frac{\sum_i^C A_i}{C} \tag{16}$$

$$\mathrm{OA} = \frac{\sum_i^C M_i}{N} \tag{17}$$

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Method	(4) 	(5) 	(6) 
	Knowledge context: The image is an aerial view of a cityscape, dominated by large buildings and streets. The tallest building in the image is the Taipei101, which is a landmark skyscraper located in Taipei, Taiwan province. It covers an area of 30,277 m^2 and shaped like a huge pagoda. It is next to Xinyi Plaza to the east, Xinyi No. 21 Park to the north. There are several cars parked in front of the building, and the streets that crisscross through these structures form a network of paths that lead to various points within this urban environment.	**Knowledge context:** The image shows a large circular building with a large square in the center. The square is People's Square of Shanghai, and the building is Shanghai Museum, located in the central area of Shanghai. The building has a large dome on top and several smaller domes on the sides. There are several trees and shrubs surrounding the building. The building appears to be made of stone or brick.	**Knowledge context:** This image shows a large airport with several runways and taxiways. The airport is Songshan Airport, located in Songshan District, Taipei, Taiwan province and serves as a domestic and regional airport. There are several vehicles parked on the tarmac, including planes and buses. The airport appears to be in use, with planes taking off and landing on the runways. The airport is surrounded by a large city with tall buildings.
GPT-4o-2024-08-06	**Object view** <Taipei 101, is a, skyscraper> <Taipei 101, is a, landmark> <Xinyi Plaza, is a, plaza> <Xinyi No. 21 Park, is a, park> **Attribute view** <Taipei 101, has area, 30,277 m^2> <Taipei 101, is shaped like, pagoda> **Spatial position view** <Taipei 101, is located in, Taipei> <Taipei, is in, Taiwan Province> **Spatial relation view** <Taipei 101, is next to, Xinyi Plaza> <Taipei 101, is next to, Xinyi No. 21 Park> <streets, crisscross through, structures> <cars, are parked in front of, Taipei 101>	**Object view** <building, is, Shanghai Museum> <square, is, People's Square of Shanghai> **Attribute view** <building, has, large dome on top> <building, has, smaller domes on sides> <building, be made of, stone or brick> **Spatial position view** <building, located in, Shanghai> **Spatial relation view** <trees and shrubs, surround, building>	**Object view** <airport, is, Songshan Airport> <airport, serves as, domestic and regional airport> **Attribute view** <Songshan Airport, has size, large> **Spatial position view** <Songshan Airport, located in, Songshan District> <Songshan District, located in, Taipei> <Taipei, located in, Taiwan Province> **Spatial relation view** <Airport, surrounded by, large city> <vehicles, parked on, tarmac> <planes, taking off and landing on, runways>
LLama-3.1-405b	**Object view** <Taipei101, is a, landmark skyscraper> **Attribute view** <Taipei101, shaped like, pagoda> <Taipei101, covers, 30,277 m^2> **Spatial position view** <Taipei101, located in, Taipei> **Spatial relation view** <Taipei101, next to, Xinyi Plaza> <Taipei101, next to, Xinyi No. 21 Park> <Cars, parked in front of, Taipei101> <Streets, crisscross through, buildings>	**Object view** <People's Square of Shanghai, is, square> <Shanghai Museum, is, building> **Attribute view** <Shanghai Museum, has, large dome> <Shanghai Museum, has, smaller domes> <building, made of, stone or brick> <building, has, large square> **Spatial position view** <Shanghai Museum, located in, Shanghai> **Spatial relation view** <trees and shrubs, surrounding, building>	**Object view** None **Attribute view** <Songshan Airport, serves as, domestic and regional airport> <city, has, tall buildings> <Songshan Airport, has, several runways and taxiways> **Spatial position view** <Songshan Airport, located in, Songshan District> **Spatial relation view** <vehicles, parked on, tarmac> <planes, taking off and landing on, runways> <airport, surrounded by, large city>
Qwen1.5-110b	**Object view** <Taipei101, is a landmark skyscraper, located in Taipei> <streets, form, a network of paths> **Attribute view** <Taipei101, covers, 30,277 m^2> <Taipei101, shaped like, a huge pagoda> **Spatial position view** None **Spatial relation view** <Taipei101, is next to, Xinyi Plaza> <Taipei101, is next to, Xinyi No. 21 Park> <streets, crisscross through, structures>	**Object view** <large square, is, People's Square of Shanghai> **Attribute view** <Shanghai Museum, has, large circular building> <Shanghai Museum, has on top, large dome> <Building, appears to be made of, stone or brick> **Spatial position view** <Shanghai Museum, is located in, Shanghai> <People's Square, is in, Shanghai> **Spatial relation view** <Shanghai Museum, surrounded by, several trees and shrubs>	**Object view** <Songshan Airport, serves as, domestic and regional airport> **Attribute view** <Songshan Airport, has feature, several runways and taxiways> <tarmac, has objects, planes and buses> **Spatial position view** <airport, located in, Songshan District> **Spatial relation view** <airport, surrounded by, large city with tall buildings> <planes, taking off/landing on, runways>
Gemini-1.5-pro-api-0514	**Object view** <Taipei 101, is a, landmark skyscraper> <Xinyi Plaza, is a, Plaza> <Xinyi No. 21 Park, is a, Park> **Attribute view** <Taipei 101, covers an area of, 30,277 m^2> <Taipei 101, shaped like, a huge pagoda> <streets, form a, network of paths> **Spatial Position view** <Taipei 101, located in, Taipei, Taiwan province> **Spatial relation view** <Taipei 101, next to, Xinyi Plaza> <Xinyi Plaza, to the east of, Taipei 101> <Xinyi No. 21 Park, to the north of, Taipei 101> <cars, parked in front of, the building>	**Object view** <Shanghai Museum, instance of, building> <People's Square, instance of, square> **Attribute view** <Shanghai Museum, has shape, circular> <Shanghai Museum, has material, stone or brick> <Shanghai Museum, has part, dome> <dome, has size, large> (referring to the main dome) <domes, has size, smaller> (referring to the side domes) <People's Square, has shape, square> **Spatial position view** <Shanghai Museum, located in, central area of Shanghai> **Spatial relation view** <Shanghai Museum, has part, People's Square> <Shanghai Museum, surrounded by, trees and shrubs>	**Object view** <Songshan Airport, is a, domestic airport> <Songshan Airport, is a, regional airport> **Attribute view** <Songshan Airport, has, several runways> <Songshan Airport, has, several taxiways> <Songshan Airport, is in, use> <Songshan Airport, is surrounded by, a large city> <city, has, tall buildings> **Spatial position view** <Songshan Airport, located in, Songshan District> <Songshan District, located in, Taipei> <Taipei, located in, Taiwan province> **Spatial relation view** <planes, parked on, tarmac> <buses, parked on, tarmac> <planes, taking off, runways> <planes, landing on, runways>

Fig. 7. *(Continued.)* Comparison of several LLMs on triplet extraction. The red text represents incorrect triplets, the blue text represents missing triplets, and the orange text represents triples with redundant information.

where N is the total number of samples in the dataset and C is the number of categories. A_i denotes the accuracy of the RSSC in the ith category, expressed as

$$A_i = \frac{M_i}{N_i} \tag{18}$$

where N_i and M_i represent the number of all optical RS images and the correctly classified images in the ith category, respectively.

As demonstrated in Table VII, the LVM-StARS performs better than the other methods do in the RSSC. It achieves higher OA and AA metrics on both the UCM and AID datasets, realizing the SOTA performance of the RSSC benchmark. In comparison, the RSMVKF demonstrates notable superiority in specific land-use scene classification tasks, with a 0.79% improvement in AA for the UCM dataset and a 1.48% improvement for the AID dataset. A visual comparison of the results between the RSMVKF and other methods is given in Table VIII. The DFAGCN yields limited benefits for classifying images from confused categories because of their inherent visual similarity and feature complexity (e.g., desert and bare land; building and dense residential; and square and

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TABLE VII
COMPARISON OF THE RSMVKF WITH SOTA RSSC METHODS

Methods	UCM		AID	
	OA	AA	OA	AA
EAM	97.76	97.76	97.18	97.18
DFAGCN	97.29	97.29	95.10	95.10
MA-FE	98.66	98.66	95.98	95.98
MaPle	95.70	95.70	92.20	92.20
LVM-StARS	99.15	99.15	97.03	97.03
TeAw	98.75	98.75	96.82	96.82
RSMVKF (ours)	**99.93**	**99.93**	**98.47**	**98.47**

park). The other compared methods exhibit slight superiority over the DFAGCN, whereas our proposed RSMVKF achieves the best performance, demonstrating an enhanced capacity to differentiate RS scenes with analogous visual features, resulting in higher accuracy in challenging scene categories of optical RS images. The results in Tables VII and VIII demonstrate the ability of the RSMVKF to discern both the broader context and finer details within images, leading to improved accuracy in RSSC. Consequently, the RSMVKF has a more pronounced advantage on the AID dataset, which is particularly challenging because of its intraclass heterogeneity and interclass homogeneity.

To explore the effects of different modules and modalities in the RSMVKF, especially the role of multiview knowledge in the performance of the RSSC task, two ablation experiments are carried out. Table IX presents the impact of the four modules in the RSMVKF, including the RS-VKG, RS-M2CL, multimodal interaction attention (MMIA), and MFS. As shown in Table IX, the RS-VKG, MMIA, and MFS contribute to improvements in the RSSC task over the use of RS-M2CL alone. Notably, the RS-VKG has the most significant effect on performance. Furthermore, Table X illustrates the effect of multimodality, particularly knowledge from OV, AV, SPV, and SRV, on the efficacy of the RSMVKF in the RSSC task. The results reveal that OV, AV, and SPV have more pronounced effects on the performance of the RSMVKF, whereas the impact of the SRV is relatively limited. This is because the information contained in OV and AV is closely related to the categories of optical RS images, serving as a supplement to the visual features of optical RS images. Moreover, SPV effectively compensates for the limitations of the visual and textual modalities in categories with distinct geographical distributions. In contrast, the enhancement provided by the SRV in the RSSC is limited.

3) RS Object Detection: The objective of RSOD is to identify and locate specific objects in optical RS images, such as buildings, vehicles, and ships. In this experiment, we focus on detecting objects in complex optical RS images to validate the effectiveness of the RSMVKF in the RSOD task. As shown in Table III, we extract highly diverse subsets of optical RS images from NWPU VHR-10 [112] and DIOR [58] as RSOD experimental datasets, which contain multiple categories of objects. These subsets are used to assess the ability of the RSMVKF to identify specific objects of interest. The mean average precision (mAP) across different target categories is used as the evaluation metric to compare the performance of the RSMVKF with that of the PISLM [133], SGPLM-IR [134], MOL [135], TSF-RGR [136], TEMO [137] and MSAM [138].

As illustrated in Table XI, the MSAM achieves SOTA performance in the RSOD task, with higher mAP values among the compared methods. Compared with TSF-RGR, the RSMVKF reveals an improvement of 3.45% in mAP on the NWPU VHR-10 dataset and 6.73% on the DIOR dataset. A visual comparison of the results between the RSMVKF and other RSOD methods is given in Table XII. In contrast to the limitations of the compared methods in fine-grained object detection when only visual and textual features are used, the RSMVKF can effectively alleviate the impact of semantic complexity and confusability in optical RS images for the RSOD task, which significantly improves the detection accuracy for objects with greater detection difficulty, e.g., bridges and overpasses. The results presented in Tables XI and XII demonstrate that the RSMVKF can effectively explore the fine-grained target information within optical RS images and optimize the performance of the model in the RSOD task by combining the visual features of multiscale optical RS images with multiview knowledge features.

To further investigate the effects of different modules and modalities on the performance of the RSMVKF in the RSOD task, we perform ablation experiments. We analyze the necessity of the RS-VKG, RS-M2CL, MMIA, and MFS methods by comparing the results of different module combinations. As shown in Table XIII, the RS-VKG, MMIA, and MFS positively influence the performance of the RSMVKF, leading to a notable increase in the mAP. Furthermore, the effects of different modalities on the RSMVKF are illustrated in Table XIV, with a particular emphasis on the contributions of OV, AV, SPV, and SRV. The results demonstrate that AV and SRV are particularly instrumental in enhancing the performance of the RSMVKF.

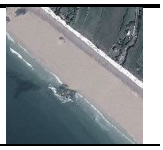
4) RS Semantic Retrieval: RSSR aims to retrieve optical RS images that satisfy textual retrieval conditions. In this experiment, we pay attention to a specific RSSR task, KNN retrieval, to verify the effectiveness of the RSMVKF. The KNN aims to retrieve the top k optical RS images that are most semantically similar to the given textual retrieval conditions. We utilize two open accessible datasets, RSICD and RSITMD, and perform sampling on them for subsequent analysis. The recall ratio $R@k$ and the mean recall ratio mR are employed as evaluation metrics for assessing the performance of the RSMVKF in comparison with other RSSR methods, including the FBCLM [139], MLT [140], and RemoteCLIP [60]. $R@k$ is denoted as

$$R@k = \frac{\mathrm{TP}@k}{\mathrm{TP}@k + \mathrm{FN}@k} \tag{19}$$

where $\mathrm{TP}@k$ represents the number of correctly retrieved images and $\mathrm{FN}@k$ denotes the number of incorrectly retrieved images. When k is set to 1, 5, or 10, the recall is calculated based on the top 1, 5, and 10 retrieval results, denoted as $R@1$, $R@5$, and $R@10$, respectively. The mR value can be

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TABLE VIII
VISUALIZATION COMPARISON OF THE RSMVKF WITH SOTA RSSC METHODS

	Input					
Output	EAM	Center	Desert	Commercial	Harbor	Beach
	DFAGCN	Airport	Bare Land	School	Harbor	Beach
	MA-FE	Airport	Bare Land	Church	Harbor	Beach
	MaPle	Airport	Desert	Commercial	Harbor	Beach
	LVM-StARS	Airport	Bare Land	School	Harbor	Beach
	TeAw	Center	Bare Land	Church	Harbor	Beach
	RSMVKF (ours)	Center	Bare Land	Commercial	Harbor	Beach
	Input					
Output	EAM	Industrial	Square	Farmland	Dense Residential	Sparse Residential
	DFAGCN	Commercial	Park	Farmland	Building	Golf Course
	MA-FE	Commercial	Park	Farmland	Dense Residential	Golf Course
	MaPle	Commercial	Park	Farmland	Dense Residential	Golf Course
	LVM-StARS	Commercial	Square	Farmland	Dense Residential	Sparse Residential
	TeAw	Industrial	Square	Farmland	Dense Residential	Golf Course
	RSMVKF (ours)	Industrial	Square	Farmland	Dense Residential	Sparse Residential
	Input					
Output	EAM	Port	Beach	Commercial	Industrial	Center
	DFAGCN	Bridge	Bridge	Square	Sparse Residential	Resort
	MA-FE	Port	Beach	Commercial	Sparse Residential	Resort
	MaPle	Port	Beach	Square	Sparse Residential	Center
	LVM-StARS	Port	Beach	Commercial	Industrial	Resort
	TeAw	Port	Beach	Commercial	Industrial	Resort
	RSMVKF (ours)	Port	Beach	Commercial	Industrial	Center
	Input					
Output	EAM	Square	Airport	Park	Center	Bridge
	DFAGCN	Commercial	Industrial	Industrial	Railway Station	Bridge
	MA-FE	Square	Airport	Park	Railway Station	Bridge
	MaPle	Commercial	Airport	Park	Center	Bridge
	LVM-StARS	Square	Airport	Park	Center	Railway Station
	TeAw	Square	Industrial	Park	Center	Railway Station
	RSMVKF (ours)	Square	Airport	Park	Center	Railway Station

expressed as

$$\mathrm{mR} = \frac{\sum_{k=1,5,10}^{N} R@k}{N} \tag{20}$$

where N represents the total number of images to be retrieved. $R@k$ and mR assume values within the range of [0, 1], with a value closer to 1 indicating superior performance.

As shown in Table XV, RemoteCLIP achieves SOTA performance among the compared RSSR methods, relying on its fine-tuning process on the CLIP model with an extended RS image dataset. The RSMVKF outperforms other RSSR methods and achieves the optimal KNN retrieval performance on both the RSITMD and RSICD datasets, as evidenced by higher $R@k$ and mR metrics. A visual comparison of the results between the RSMVKF and other RSSR methods on the RSITMD dataset is given in Table XVI. The results show that the FBCLM and MLT exhibit limitations in discerning spatial relationships between entities in optical RS images. For example, the accuracy of retrieving optical RS images satisfied the given conditions: "two baseball fields over against each other surrounded by many trees" and "a round pond surrounded by dense vegetation" are insufficient. In contrast, the RSMVKF, by integrating knowledge from the four perspectives, can discern more subtle semantic details in optical

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TABLE IX
RESULTS OF ABLATION MODEL STUDIES ON RSSC

Ablation Model				UCM		AID	
RS-VKG	RS-M2CL	MMIA	MFS	OA	AA	OA	AA
×	√	×	×	99.21	99.21	97.28	97.28
√	√	×	×	99.65	99.65	97.45	97.45
√	√	√	×	99.71	99.71	98.03	98.03
√	√	√	√	**99.79**	**99.79**	**98.37**	**98.37**

TABLE X
RESULTS OF ABLATION MODAL STUDIES ON RSSC

Ablation Modal						UCM		AID	
IMG	Text	KG				OA	AA	OA	AA
		OV	AV	SPV	SRV				
√	√	√	×	×	×	99.57	99.57	97.45	97.45
√	√	×	√	×	×	99.64	99.64	97.55	97.55
√	√	×	×	√	×	99.67	99.67	98.22	98.22
√	√	×	×	×	√	99.76	99.76	98.32	98.32
√	√	√	√	√	√	**99.83**	**99.83**	**98.47**	**98.47**

TABLE XI
COMPARISON OF THE RSMVKF WITH SOTA RSOD METHODS

Methods	NWPU VHR-10 mAP	DIOR mAP
PISLM	63.8	28.6
SGPLM-IR	65.2	28.5
MOL	75.46	29.21
TSF-RGR	74.52	28.97
TEMO	76.23	29.35
MSAM	77.45	29.72
RSMVKF (ours)	**80.12**	**31.72**

RS images. This capability effectively addresses the issue of low retrieval accuracy arising from the inherent semantic intricacy and spatial complexity of optical RS images. Consequently, it enhances the model's performance in the RSSR task, particularly in KNN retrieval.

To investigate the effect of different modules of the RSMVKF, as well as the influence of different types of view knowledge on the model's performance, ablation experiments are carried out. Table XVII shows the necessity of the key modules, i.e., RS-VKG, RS-M2CL, MMIA, and MFS, in the RSSR task. All the modules play an important role in enhancing the performance of the RSMVKF, and the complete structured the RSMVKF yields improvements in the mR for the RSSR task of 1.98% and 2.62%, respectively. Table XVIII further illustrates the effects of OV, AV, SPV, and SRV in the RSSR task. The results show that in most cases, the RSMVKF with the simultaneous introduction of multiview knowledge exhibits better performance than does engaging single-view knowledge.

5) RS Image Captioning: This task involves generating a textual description for a given RS image that accurately reflects its main features and contents. In this experiment, spatial semantic description is selected as the target of specific RSCP. The focus is on analyzing the description accuracy of the RSMVKF in terms of object categories, spatial locations, and spatial relations, thereby evaluating the method's overall performance. We use the RSICD [113] and UCM-Caption [115] datasets for the experiment and employ common metrics such as BLEU [141], METEOR [142], CIDEr [143], and ROUGE_L [144] to validate the effectiveness of the RSMVKF in this specific RS image caption. BLEU, which was originally developed for machine translation, measures the similarity between the generated text and the reference text by dividing them into n-grams and applying a penalty on the basis of the length difference between the texts [141]

$$\text{BLEU} = \text{BP} \times \exp\left(\sum_{n=1}^{N} W_n \times \log P_n\right). \tag{21}$$

Here, P_n represents the precision of the n-grams, and W_n denotes the weight of the n-grams. BP is the penalty factor that accounts for the difference in text length and can be expressed as

$$\text{BP} = \begin{cases} 1, & l_c > l_s \\ \exp\left(1 - \dfrac{l_r}{l_c}\right), & l_c \le l_s \end{cases} \tag{22}$$

where l_c represents the length of the generated image caption and l_s is the length of the closest match in the reference text. METEOR is a metric that is based on the weighted harmonic mean of precision and recall and focuses on single-word precision and recall. It accounts for multiple levels of lexical, syntactic, and semantic matching between the candidate text and the reference text, utilizing precision and recall across the entire corpus to calculate evaluation results [142]. METEOR can be expressed as

$$\text{METEOR} = (1 - \text{Pen}) \times F_{\text{mean}} \tag{23}$$

where Pen denotes the penalty term and F_{mean} represents the harmonic mean of precision and recall. Pen and F_{mean} can be expressed as

$$\text{Pen} = \gamma \left(\frac{\text{ch}}{m}\right)^{\theta} \tag{24}$$

$$F_{\text{mean}} = \frac{P_m R_m}{\alpha P_m + (1-\alpha) R_m} \tag{25}$$

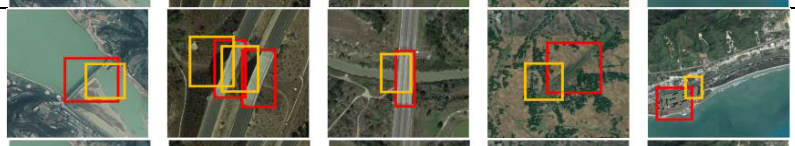
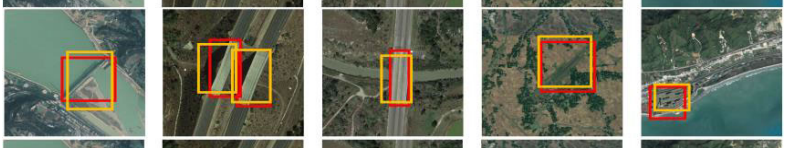
where α, γ, and θ are the default parameters used for evaluation, m represents the number of unigrams, and ch denotes the number of consecutively ordered chunks in the generating sentence. P_m denotes the precision, and R_m denotes the recall, which is computed as

$$P_m = \frac{|m|}{\sum_k h_k(c_i)}. \tag{26}$$

CIDEr accounts for the lexical diversity and consistency between the generated text and the reference text. It uses multiple reference texts to calculate the weight of each n-gram, with n-grams that occur more frequently across different reference texts corresponding to higher weights [143]. CIDEr

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TABLE XII
VISUALIZATION COMPARISON OF THE RSMVKF WITH SOTA RSOD METHODS

Input		
Output	PISLM	
	SGPLM-IR	
	MOL	
	TSF-RGR	
	TEMO	
	MSAM	
	RSMVKF (ours)	
Input		
Output	PISLM	
	SGPLM-IR	
	MOL	
	TSF-RGR	

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TABLE XII
(Continued.) VISUALIZATION COMPARISON OF THE RSMVKF WITH SOTA RSOD METHODS

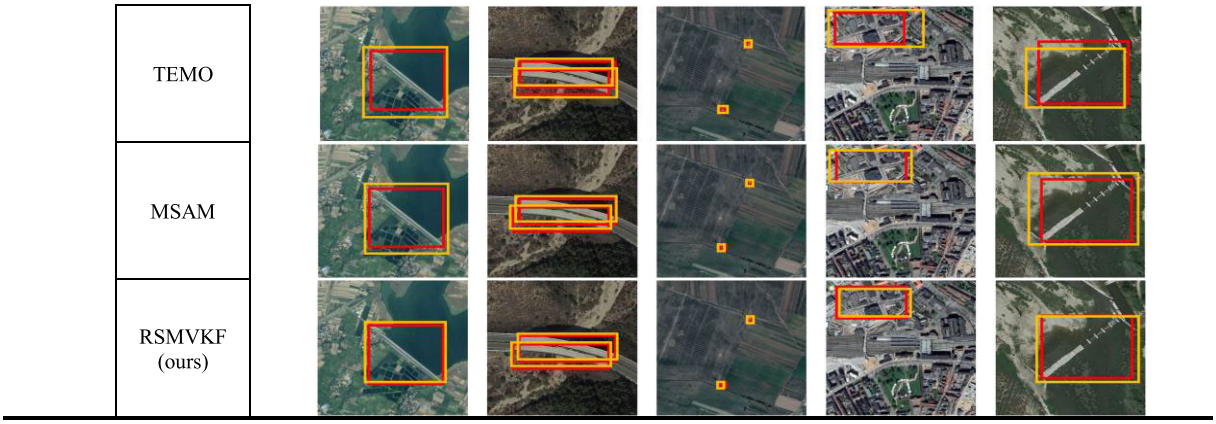


TABLE XIII
RESULTS OF ABLATION MODEL STUDIES ON RSOD

Ablation Model				NWPU VHR-10	DIOR
RS-VKG	RS-M2CL	MMIA	MFS	mAP	mAP
×	√	×	×	74.26	28.76
√	√	×	×	78.37	30.22
√	√	√	×	78.30	32.09
√	√	√	√	**79.12**	**33.36**

TABLE XIV
RESULTS OF ABLATION MODAL STUDIES ON RSOD

Ablation Modal						NWPU VHR-10	DIOR
IMG	Text	KG				mAP	mAP
		OV	AV	SPV	SRV		
√	√	√	×	×	×	75.43	28.95
√	√	×	√	×	×	78.66	30.43
√	√	×	×	√	×	75.97	29.81
√	√	×	×	×	√	78.35	32.52
√	√	√	√	√	√	**79.25**	**33.38**

TABLE XV
COMPARISON OF THE RSMVKF WITH SOTA RSSR METHODS

Methods	RSICD				RSITMD			
	R@1	R@5	R@10	mP	R@1	R@5	R@10	mP
FBCLM	10.44	37.01	57.94	35.13	13.54	38.74	56.94	36.41
MLT	17.61	49.73	66.59	44.64	9.14	28.96	44.59	27.56
RemoteCLIP	23.76	59.51	74.73	52.67	14.73	39.93	56.58	37.08
RSMVKF (ours)	**25.67**	**61.34**	**78.98**	**56.93**	**18.74**	**43.56**	**59.35**	**40.55**

is computed as

$$\mathrm{CIDEr}(c_i, S_i) = \sum_{n=1}^{N} \omega_n \mathrm{CIDEr}_n(c_i, S_i) \tag{27}$$

$$\mathrm{CIDEr}_n(c_i, S_i) = \frac{1}{M} \sum_{j=1}^{M} \frac{g^n(c_i) \times g^n(s_{ij})}{\|g^n(c_i)\| \times \|g^n(s_{ij})\|} \tag{28}$$

$$g_k(c_i) = \mathrm{TF}(k) \times \mathrm{IDF}(k) \tag{29}$$

where c_i represents the caption corresponding to the ith RS image, s_{ij} is the jth reference text among multiple reference texts associated with the RS image, M is the number of reference texts for each RS image, and k refers to the kth n-gram in the sentence. ROUGE-L [144] measures how many n-grams from the reference texts appear in the output, focusing on the longest common subsequence between the output text and the reference text. ROUGE-L is defined as

$$R_{\mathrm{lcs}} = \frac{\mathrm{LCS}(X, Y)}{m} \tag{30}$$

$$P_{\mathrm{lcs}} = \frac{\mathrm{LCS}(X, Y)}{n} \tag{31}$$

$$\mathrm{ROUGE_L} = F_{\mathrm{lcs}} = \frac{(1+\beta^2) R_{\mathrm{lcs}} P_{\mathrm{lcs}}}{R_{\mathrm{lcs}} + \beta^2 P_{\mathrm{lcs}}} \tag{32}$$

where R_{lcs} represents the recall, P_{lcs} denotes the precision, $\mathrm{LCS}(X, Y)$ is the length of the longest common subsequence between X and Y, and $\beta = P_{\mathrm{lcs}}/R_{\mathrm{lcs}}$.

We compare the RSMVKF with three RSCP methods: RSGPT [28], SkyEyeGPT [145], and RS-CapRet [30]. Among these methods, RSGPT is a pretrained model on the specialized RSCP dataset RSICap [28], whereas the SkyEyeGPT achieves SOTA results in RSCP at the time of its proposal. The comparison results are presented in Table XIX. As shown in the results, for the specific task of spatial semantic description in RS images, our model outperforms the aforementioned methods across most metrics. For the RSICD dataset, the RSMVKF outperforms the RSGPT in terms of all the metrics and outperforms the SkyEyeGPT in terms of most of the metrics. Notably, the METEOR metric improves by 10.73%, and the ROUGE_L metric improves by 9.13%. For the UCM-Captions dataset, the RSMVKF outperforms both the RSGPT and RS-CapRet methods across all the metrics and achieves performance comparable to that of the SkyEyeGPT in terms of the BLEU metrics, with improvements in the METEOR and ROUGE_L metrics, where the METEOR metric improves by 8.87%. Fig. 8 presents the experimental results of image captioning on the RSICD dataset, comparing the RSMVKF with other methods. In Fig. 8, the missing information types in the comparison methods are highlighted in red. The results indicate that the RSGPT and RS-CapRet methods summarize the scene and describe the entities and their quantities within the image. However, they omit details regarding spatial relations and attribute information, such as shape and color. While the SkyEyeGPT can describe entity categories and simple spatial

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TABLE XVI
VISUALIZATION COMPARISON OF THE RSMVKF WITH SOTA RSSR METHODS

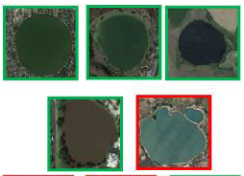
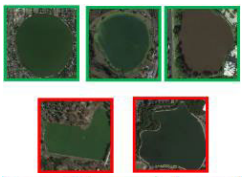
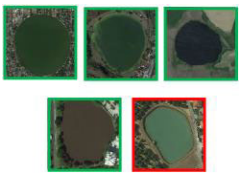
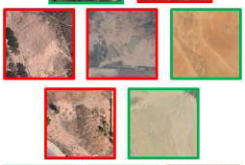
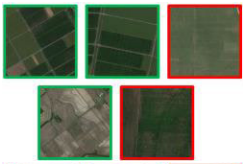
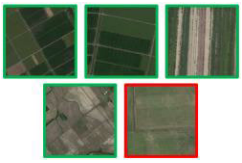
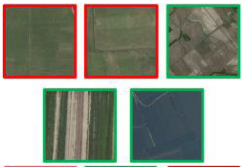
Input	Output			
	FBCLM	MLT	RemoteCLIP	RSMVKF (ours)
Two baseball fields over against each other surrounded by many trees.				
A round pond surrounded by dense vegetation.				
Deserts with a few green drought-tole-rant plants				
Bridges connecting areas with vegetation				
Pictures of farmland				
Parks with vegetation or water bodies				

TABLE XVII
RESULTS OF ABLATION MODEL STUDIES ON RSSR

Ablation Model				RSITMD				RSICD			
RS-VKG	RS-M2CL	MMIA	MFS	R@1	R@5	R@10	mR	R@1	R@5	R@10	mR
×	√	×	×	18.37	50.19	62.98	43.85	11.45	32.98	49.07	31.17
√	√	×	×	20.14	53.18	67.37	47.90	13.87	35.87	52.34	34.03
√	√	√	×	23.46	57.32	72.19	50.99	15.34	39.56	55.89	36.93
√	√	√	√	25.65	61.22	78.76	55.21	18.68	43.42	59.31	40.23

relationships between entities, it cannot provide a comprehensive description of shape and color information. In contrast, the RSMVKF model offers a more detailed and comprehensive description of the entities, including their attributes and spatial information, demonstrating significantly better performance than previous SOTA models do, particularly in the task of spatial semantic description.

To investigate the impact of different components of the RSMVKF with knowledge from different views on the performance of the RSCP, two sets of ablation experiments are conducted. The necessity of the components, namely, RS-VKG, RS-M2CL, MMIA, and MFS, in the proposed approach, is analyzed in Table XX by comparing the performance of the RSMVKF in the RSCP via different component combinations. The results in Table XIX indicate that the RS-VKG, RS-M2CL, MMIA, and MFS each play crucial roles in the performance of the RSMVKF, with improvements observed in the BLEU, METEOR, ROUGE_L, and CIDEr metrics for the RSCP. Table XXI presents the effects of single-view and multiview combinations of OV, AV, SPV, and SRV on the performance of the RSCP. The results show that OV, AV, and SPV each contribute to a significant improvement in

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TABLE XVIII
RESULTS OF ABLATION MODAL STUDIES ON RSSR

Ablation Modal						RSITMD				RSICD			
IMG	Text	KG				R@1	R@5	R@10	mR	R@1	R@5	R@10	mR
		OV	AV	SPV	SRV								
√	√	√	×	×	×	24.51	60.21	75.34	53.35	15.67	41.34	56.78	37.93
√	√	×	√	×	×	24.55	60.41	75.53	53.41	15.71	41.44	56.91	38.04
√	√	×	×	√	×	24.76	60.60	75.90	53.78	16.04	41.81	57.38	38.55
√	√	×	×	×	√	25.17	61.05	76.90	54.29	16.27	42.75	58.12	39.48
√	√	√	√	√	√	25.67	61.34	78.98	55.33	18.67	43.04	58.21	40.34

TABLE XIX
COMPARISON OF THE RSMVKF WITH SOTA RSCP METHODS

Dataset	Methods	BLEU1	BLEU2	BLEU3	BLEU4	METEOR	ROUGE_L	CIDEr
RSCID	RSGPT	0.703	0.542	0.44	0.368	0.301	0.533	1.029
	SkyEyeGPT	0.741	0.622	0.529	0.455	0.376	0.649	2.605
	RS-CapRet	0.867	0.767	0.673	0.6	0.354	0.626	0.837
	RSMVKF (ours)	**0.882**	**0.771**	**0.684**	**0.583**	**0.412**	**0.681**	**2.621**
UCM-Captions	RSGPT	0.861	0.791	0.723	0.657	0.422	0.783	3.332
	SkyEyeGPT	0.833	0.76	0.699	0.645	0.447	0.786	3.429
	RS-CapRet	0.907	0.857	0.816	0.784	0.462	0.795	2.368
	RSMVKF (ours)	**0.911**	**0.862**	**0.813**	**0.788**	**0.523**	**0.81**	**3.435**

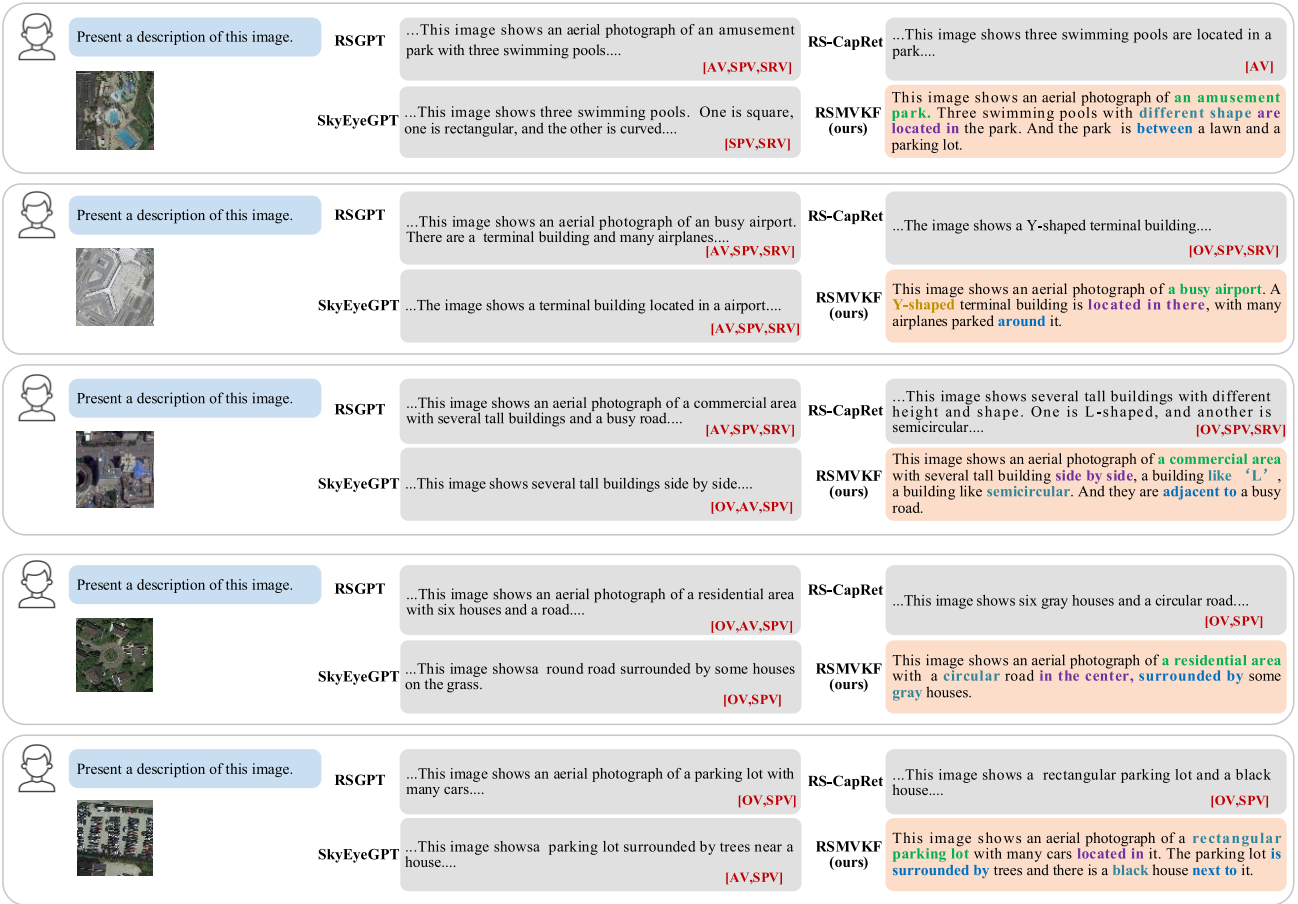


Fig. 8. Visualization comparision of the RSMVKF with SOTA RSCP methods. The OV, AV, SPV, and SRV descriptions are highlighted in green, yellow, purple, and blue, respectively.

model performance. The model's performance with multiview knowledge is notably superior to the results obtained using only single-view knowledge on both the RSCID and UCM-Captions datasets.

6) RS Visual Grounding: RSVG predicts the location of a specific object in an optical RS image via a language expression, which necessitates the capacity of alignment and fusion of features in different modalities. We take RSVGD [116]

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TABLE XX
RESULTS OF ABLATION MODEL STUDIES ON RSCP

Ablation Model				RSICD							UCM Caption						
RS-VKG	RS-M2CL	MMIA	MFS	BLEU1	BLEU2	BLEU3	BLEU4	METEOR	ROUGE_L	CIDEr	BLEU1	BLEU2	BLEU3	BLEU4	METEOR	ROUGE_L	CIDEr
×	√	×	×	0.642	0.609	0.517	0.401	0.294	0.549	1.843	0.694	0.652	0.601	0.582	0.343	0.529	2.516
√	√	×	×	0.767	0.652	0.578	0.455	0.331	0.561	2.012	0.765	0.701	0.673	0.629	0.392	0.582	2.875
√	√	√	×	0.801	0.726	0.623	0.534	0.374	0.642	2.432	0.818	0.768	0.742	0.694	0.488	0.637	3.012
√	√	√	√	0.882	0.771	0.684	0.583	0.392	0.681	2.621	0.911	0.862	0.813	0.788	0.503	0.810	3.435

TABLE XXI
RESULTS OF ABLATION MODAL STUDIES ON RSCP

Ablation Model						RSICD							UCM						
IMG	Text	KG				BLEU1	BLEU2	BLEU3	BLEU4	METEOR	ROUGE_L	CIDEr	BLEU1	BLEU2	BLEU3	BLEU4	METEOR	ROUGE_L	CIDEr
		OV	AV	SPV	SRV														
√	√	√	×	×	×	0.812	0.671	0.632	0.566	0.377	0.618	2.332	0.871	0.821	0.781	0.722	0.479	0.652	3.077
√	√	×	√	×	×	0.794	0.659	0.633	0.564	0.382	0.591	2.122	0.862	0.818	0.773	0.717	0.468	0.627	2.794
√	√	×	×	√	×	0.763	0.641	0.616	0.489	0.375	0.514	1.82	0.852	0.781	0.754	0.704	0.433	0.548	2.751
√	√	×	×	×	√	0.834	0.711	0.671	0.569	0.388	0.629	2.574	0.879	0.835	0.795	0.756	0.494	0.678	3.162
√	√	√	√	√	√	0.882	0.771	0.684	0.583	0.392	0.681	2.621	0.911	0.862	0.813	0.788	0.503	0.81	3.435

TABLE XXII
COMPARISON OF THE RSMVKF WITH SOTA RSVG METHODS

Dataset	Methods	Pr@0.5	Pr@0.6	Pr@0.7	Pr@0.8	Pr@0.9	meanIoU	cumIoU
RSVGD	MSAM	62.92	59.54	51.23	36.32	12.82	54.24	67.21
	MSCMA	74.54	70.10	65.16	54.91	32.54	63.10	72.19
	LPVA	73.21	68.26	63.31	49.01	27.45	62.65	75.32
	RSMVKF (ours)	77.32	70.82	67.50	56.97	38.38	69.41	78.50
DIOR-RSVG	MSAM	74.23	69.01	61.32	49.04	24.26	64.88	77.13
	MSCMA	83.41	81.03	75.91	65.52	43.53	74.02	82.22
	LPVA	82.27	77.44	72.25	60.98	39.55	72.35	85.11
	RSMVKF (ours)	85.52	83.18	79.06	69.67	47.39	78.16	88.25

and DIOR-RSVG [117] as experimental datasets and extract images containing a variety of geographic entities as subsets to evaluate the performance of the RSMVKF in the area-specific-oriented RSVG task. Referring to [117], we adopt Pr@0.5–0.9, meanIoU, and cumIoU as evaluation metrics. Given an RSVG query pair, a prediction is deemed to be accurate if the Intersection over Union (IoU) between the predicted result and the ground-truth bounding box exceeds the predefined threshold. The IoU thresholds are set to 0.5, 0.6, 0.7, 0.8, and 0.9; thus, the corresponding accuracy metrics are Pr@0.5, Pr@0.6, Pr@0.7, Pr@0.8, and Pr@0.9. Consistent with [117], meanIoU and cumIoU are calculated as

$$\text{meanIoU} = \frac{1}{M}\sum_{i}\frac{\text{IA}_i}{\text{UA}_i} \tag{33}$$

$$\text{cumIoU} = \frac{\sum_i \text{IA}_i}{\sum_i \text{UA}_i} \tag{34}$$

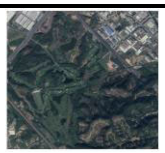
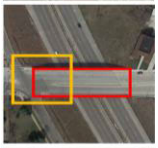
where M denotes the size of the dataset, and IA_i and UA_i denote the areas of intersection and union regions between the prediction result and the actual labeled bounding box of the ith sample pair, respectively.

We compare the RSMVKF and other RSVG methods, including the MSAM [138], MSCMA [146], and LPVA [147], as shown in Table XXII. The results demonstrate that MSCMA outperforms the other methods in terms of the majority of the metrics. Conversely, the RSMVKF is the most effective method for addressing area-specific-oriented RSVG. For the RSVG dataset, the RSMVKF achieves a meanIoU of 69.41% and a cumIoU of 78.50%, representing improvements of 7.6% and 5.3% over the MSCMA, respectively. For the DOIR-RSVG dataset, the RSMVKF results in improvements of 9.8% and 8.7%. These findings highlight the ability of the RSMVKF method to capture fine-grained object information effectively, as well as the spatial position and spatial relation information between objects in optical RS images, thus improving the performance of the model in the RSVG task in a specific area. The visual comparison results are presented in Table XXIII, where the predicted outcomes of various methods are indicated in yellow boxes and the ground-truth values are marked in red. The findings illustrate that the RSMVKF is more effective than other methods at identifying and locating objects in optical RS images containing intricate features, benefiting from its proficiency in discerning and interpreting fine-grained semantics. For example, the RSMVKF can accurately locate a golf course in a complex optical RS image that also includes other objects such as buildings and bridges.

Two subsequent ablation experiments are conducted to further investigate the role of different modules and different view knowledge on the performance of the RSMVKF in the area-specific-oriented RSVG task. Table XXIV presents a comparative analysis with different combinations of modules. Compared with the RS-M2CL-only approach, the RS-VKG,

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TABLE XXIII
VISUALIZATION COMPARISON OF THE RSMVKF WITH SOTA RSVG METHODS

Input	Image					
	Query	A gray overpass in the middle	The gray bridge in the middle	A windmill in the middle	Golf field in the image	The train station in the middle
Output	MSAM					
	MSCMA					
	LPVA					
	RSMVKF (ours)					
Input	Image					
	Query	The cyan and gray dam	A gray large airport in the middle	The large long airport	The dam on the top	The train station in the middle
Output	MSAM					
	MSCMA					
	LPVA					
	RSMVKF (ours)					

MMIA, and MFS significantly increase the RSVG accuracy. The most optimal performance in the RSVG task is achieved when all four modules are integrated into the RSMVKF. Furthermore, Table XXV details the effects of OV, AV, SPV, and SRV on model performance. The results demonstrate that OV, AV, and SPV lead to pronounced improvements. Compared with that of single-view knowledge, the performance of the RSMVKF with multiview knowledge in the RSVG task is considerably better on both the RSVGD and DIOR-RSVG datasets. The information of the object category, attribute, and

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TABLE XXIV
RESULTS OF ABLATION MODEL STUDIES ON RSVG

Ablation Model				RSVGD		DIOR-RSVG	
RS-VKG	RS-M2CL	MMIA	MFS	Pr@0.5	meanIoU	Pr@0.5	meanIoU
×	√	×	×	75.63	67.32	66.64	57.23
√	√	×	×	81.48	72.43	72.32	61.23
√	√	√	×	85.11	75.56	74.76	64.44
√	√	√	√	85.67	77.90	76.74	69.06

TABLE XXV
RESULTS OF ABLATION MODAL STUDIES ON RSVG

Ablation Modal						RSVGD		DIOR-RSVG	
IMG	Text	KG				Pr@0.5	meanIoU	Pr@0.5	meanIoU
		OV	AV	SPV	SRV				
√	√	×	×	×	×	74.54	67.61	64.16	57.69
√	√	√	×	×	×	77.77	70.62	67.26	60.74
√	√	×	√	×	×	78.77	71.79	68.56	63.50
√	√	×	×	√	×	79.22	72.61	69.04	63.56
√	√	×	×	×	√	81.73	75.15	72.02	66.49
√	√	√	√	√	√	84.68	77.76	74.79	68.98

TABLE XXVI
RESULTS OF ABLATION MODEL STUDIES ON RSVQA

Ablation Model				RSVQA-HR	RSVQA-LR
RS-VKG	RS-M2CL	MMIA	MFS	AA	AA
×	√	×	×	0.795	0.817
√	√	×	×	0.821	0.851
√	√	√	×	0.875	0.883
√	√	√	√	**0.922**	**0.938**

TABLE XXVII
RESULTS OF ABLATION MODAL STUDIES ON RSVQA

Ablation Modal						RSVQA-HR	RSVQA-LR
IMG	Text	KG				AA	AA
		OV	AV	SPV	SRV		
√	√	√	×	×	×	0.734	0.739
√	√	×	√	×	×	0.752	0.785
√	√	×	×	√	×	0.703	0.721
√	√	×	×	×	√	0.854	0.873
√	√	√	√	√	√	**0.922**	**0.938**

spatial relation provided by OV, AV, and SRV, respectively, serve as valuable supplements to visual and textual features when facing the semantic complexity and confusability of optical RS images.

7) RS Visual Question Answering: This task involves answering natural language questions related to RS images by understanding and reasoning about the images. In this experiment, we selected region-specific VQA as a specific RSVQA to compare the performance of the RSMVKF with that of other methods. We utilized the publicly available RSVQA-LR and RSVQA-HR datasets [118] as representative datasets, selecting images with diverse entity categories and high analytical complexity to construct the experimental dataset. The AA is chosen as the evaluation metric to compare the performance of the RSMVKF with that of other SOTA methods. AA is a commonly used metric in RSVQA; it measures performance by calculating the quantitative relationship between generated responses and multiple valid answers. The values of AA range from 0 to 1; a value closer to 1 indicates better performance.

The RSMVKF is compared with four generic VLMs, namely, MiniGPT-4 [17], MiniGPTv2 [148], Qwen-VL [19], and LLaVA-1.5 [119], as well as RS domain VLMs, i.e., RSGPT [28], GeoChat [29], and SkyEyeGPT [145]. The results are shown in Fig. 9. An analysis of these results indicates that the RSMVKF performs exceptionally well in region-specific RSVQA, demonstrating significant performance advantages over the comparison methods. Specifically, for the RSVQA-HR dataset, the RSMVKF outperforms the SOTA professional model RSGPT, achieving an AA of 92.23%. For the RSVQA-LR dataset, the RSMVKF significantly outperforms the generic VLMs, and the RSMVKF's AA is 11.38% higher than that of the SkyEyeGPT and is comparable to that of the SOTA method, the RSGPT. Fig. 10 provides a visualization of the comparison results of the RSMVKF with three representative methods, namely, RSGPT, GeoChat, and SkyEyeGPT, on the RSVQA-LR dataset. The types of missing information in the comparison methods are highlighted in red. The analysis shows that the RSMVKF method significantly outperforms the comparison methods across three types of questions: scene classification, object presence, and spatial relation. Among the comparison methods, only the SkyEyeGPT addresses some questions on scene classification and object presence, but it performs poorly on questions related to spatial relations. The RSMVKF, by integrating multiview knowledge, provides more accurate answers and delivers richer results that encompass information from various views.

We conducted ablation experiments to investigate the effects of different components of the RSMVKF, each incorporating various types of view knowledge, on the performance of the RSVQA. Table XXVI presents the performance of the RSMVKF with different combinations of the RS-VKG, RS-M2CL, MMIA, and MFS. The results indicate that all these components significantly contribute to the performance of the RSMVKF. Table XXVII lists the effects of single and multiview combinations of OV, AV, SPV, and SRV on RSVQA. The results demonstrate that the RSMVKF with multiview knowledge outperforms the model using only single-view knowledge on both the RSVQA-HR and RSVQA-LR datasets. Specifically, the model's AA is lowest when only SPV is used, with scores of 70.3% and 72.1%, respectively. Introducing OV or AV alone results in modest improvements, with AV providing greater enhancement than OV. Notably, the model's performance significantly improves when SRV is incorporated alone, highlighting the crucial role of SRV in enhancing the model's knowledge discovery capabilities.

D. Parameter Analysis

The outcomes of a model are strongly influenced by hyperparameter values. Therefore, we conduct parameter analysis experiments to confirm the impact of these hyperparameters and determine their optimal values. We focus on the key hyperparameters, λ_{ITC}, λ_{IVC}, λ_{ITM}, and λ_{ITA}. Among them, λ_{ITC} and λ_{IVC} are used for balancing IIC, ITC, and IVC losses, whereas λ_{ITKM} and λ_{ITKA} are used for balancing ITM and

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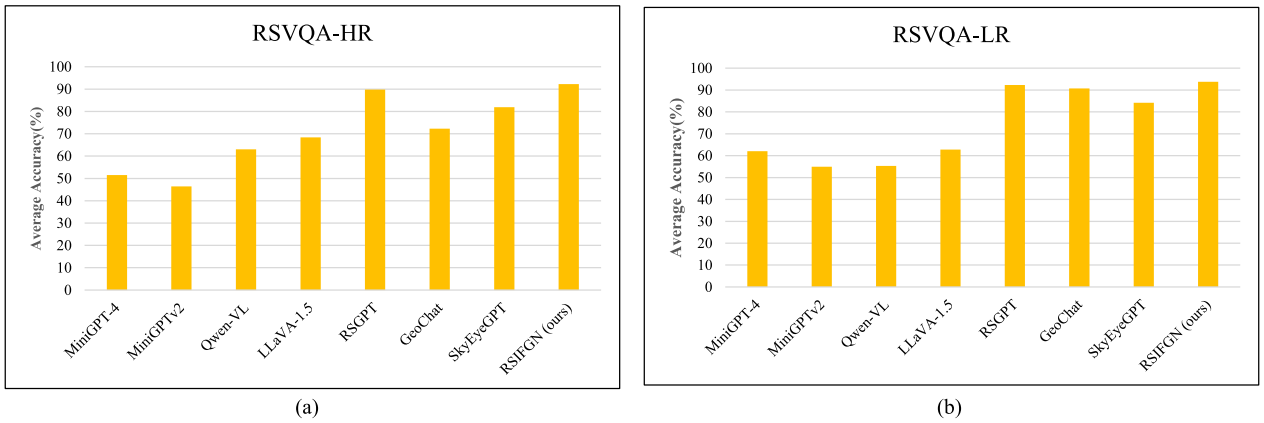


Fig. 9. Comparison of the RSMVKF with different SOTA RSVQA methods. (a) RSVQA-HR. (b) RSVQA-LR.



Fig. 10. Visualization comparation of the RSMVKF with SOTA RSVQA methods on the RSVQA-LR dataset.

ITA losses. In the experiments, λ_{ITC} and λ_{IVC} increase from 0.1 to 0.9 in incremental steps. The corresponding results are given in Fig. 11(a), taking $R@10$ for the RSSR task as the evaluation metric. The ITC loss significantly affects the MGCL loss, and the results show that the RSMVKF achieves optimal performance when λ_{ITC} is set to 0.35. Additionally, the IVC loss further improves the performance of the RSMVKF, with the optimal value of λ_{IVC} being approximately 0.55.

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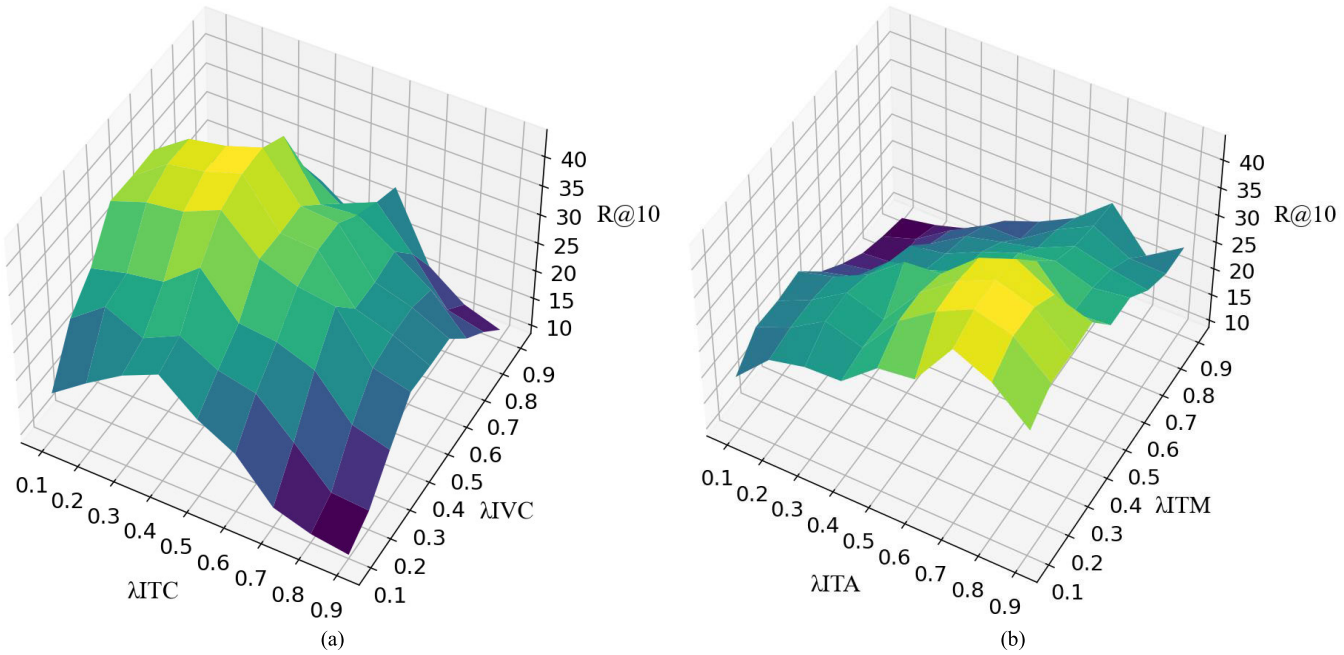


Fig. 11. $R@10$ of loss function parameter experiment on RSSR task on RSITMD dataset. (a) Impact of hyperparameters λ_{ITC} and λ_{IVC}. (b) Impact of hyperparameters λ_{ITA} and λ_{ITM}.

The values of λ_{ITKM} and λ_{ITKA} are varied within the range of [0.1, 0.9] with an increment of 0.1. The corresponding results are given in Fig. 11(b). The ITM loss significantly influences the MMFL, and the results indicate that the RSMVKF exhibits optimal performance on the RSSR task when λ_{ITKM} is approximately 0.75. Furthermore, ITA loss contributes to the MMFL and influences the performance of the RSMVKF. The RSMVKF attains its optimal performance when λ_{ITA} assumes a value of approximately 0.25.

V. Discussion

In future research, additional challenges merit further consideration in terms of the practical applications of our research. First, we use an LLM for view-level knowledge graph construction and knowledge triplet extraction, which inevitably suffers from incompleteness or incorrectness. In this study, we take several measures to address these issues, including conducting few-shot in-context learning on the LLM, inserting processing modules between the output of the LLM and the subsequent knowledge encoder, and adjusting the temperature parameter during model inference. Despite the excellent performance of these methods, a fully end-to-end RS representation learning approach cannot yet be provided in this article due to the inability to ensure consistent learning progress between the two stages of the method. In the future, we plan to encapsulate these methods and their dependent rules into a workflow service. By enhancing the collaboration between the data flow and the control flow of the service, the overall efficiency can be further improved. In addition, there is external information outside the four human-like cognition views that merit attention. For example, the evolution of features in the time dimension and the intrinsic relationships among them in terms of semantic relationships and spatial–temporal relationships are important. In future research, on the one hand, we will extend the analysis dimensions to explore semantics in optical RS images of greater profundity and diversity, and integrate them into the existing view-level knowledge graph. Moreover, we should strive to expand the scope of view knowledge and provide more diverse and flexible views for optical RS image comprehension. On the other hand, we investigate preferable solutions for the organization and correlation of view-level knowledge to discover the implicit semantics among multisource optical RS images.

VI. Conclusion

In this article, we aim to address the problems of existing research methods in the comprehension and application of the fine-grained semantics and relationships implied in optical RS images. We concentrate on the structured representations of external knowledge from different views of human-like cognition and leverage these representations to enhance the model's comprehensive semantic understanding capability. We design a semantic-augmented the RSMVKF approach with two stages, which combines a view-level knowledge graph with an asymmetric RS-M2CL. Compared with other SOTA methods, we validate the effectiveness of the RSMVKF in downstream perception and reasoning tasks.

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Lilu Zhu received the M.Sc. degree from Guangxi Normal University, Guilin, China, in 2015, and the Ph.D. degree from the School of Information Science and Technology, University of Science and Technology of China, Hefei, China, in 2023.

He is currently an Associate Professor with Suzhou Aerospace Information Research Institute, Suzhou, China. His main research interests include geospatial data service and distributed system architecture.



Xiaolu Su received the M.Sc. degree from the School of Microelectronics Engineering, Southeast University, Nanjing, China, in 2017.

She is currently an Assistant Researcher with Suzhou Aerospace Information Research Institute, Suzhou, China. Her main research interests include geospatial data organization and service.



Jiaxuan Tang received the B.Sc. degree from China University of Geosciences, Beijing, China, in 2018, and the Ph.D. degree from the University of Chinese Academy of Sciences, Beijing, in 2023.

She is currently an Assistant Researcher with Suzhou Aerospace Information Research Institute, Suzhou, China. Her research interests include geospatial data service and understanding.



Yanfeng Hu received the B.S. degree from Xidian University, Xi'an, China, in 1999, and the Ph.D. degree from the Institute of Electronics, Chinese Academy of Sciences, Beijing, China, in 2005.

He is currently a Professor with the Aerospace Information Research Institute, Chinese Academy of Sciences. His research interests include natural language processing and remote sensing image understanding.



Yang Wang (Member, IEEE) received the B.Sc. degree from Beihang University, Beijing, China, in 2007, and the M.Sc. and Ph.D. degrees from Peking University, Beijing, in 2010 and 2014, respectively.

She is currently an Associate Professor with Suzhou Aerospace Information Research Institute, Suzhou, China. Her research interests include geospatial data organization and visualization.

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